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PHYSICS

1) Two metals plates are placed in parallel, capacitance of capacitor is C then what will be potential energy stored in the region between two plates :-



(1) $\frac{1}{2}\frac{Q^2}{C}$

(2) $\frac{9}{2}\frac{Q^2}{C}$

(3) $\frac{3}{2}\frac{Q^2}{C}$

(4) zero

2) A and B are concentric conducting spherical shells. A is given a positive charge while B is earthed. Then :-



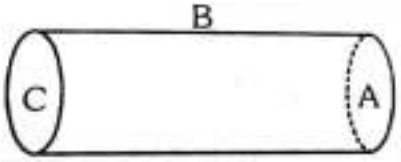
(1) A and B both will have the same charge densities

(2) the potential inside A and outside B will zero

(3) the electric field between A and B is non zero

(4) the electric field inside A and outside B is non zero.

3) A hollow cylinder has a charge q coulombs located within it symmetrically. If ϕ is the electric flux in units of volt-meter associated with the curved surface B, the flux linked with the plane surface A in units of volt-meters will be :-



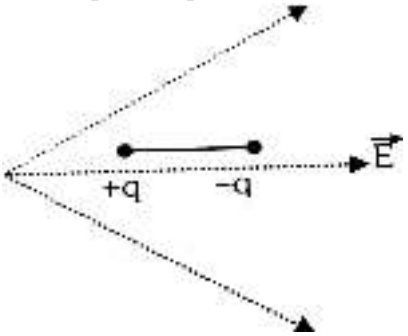
(1) $\frac{q}{\epsilon_0} - \phi$

(2) $\frac{1}{2}\left[\frac{q}{\epsilon_0} - \phi\right]$

(3) $\frac{q}{2\epsilon_0}$

(4) $\frac{\phi}{3}$

4) A dipole is placed in an electric field as shown. In which direction will it move?



(1) towards the left as its potential energy will increase.

(2) towards the right as its potential energy will decrease.

(3) towards the left as its potential energy will decrease.

(4) towards the right as its potential energy will increase.

5) A dipole of dipole moment 'p' is placed in a non-uniform electric field along x-axis. Electric field is increasing at the rate of 1 V/m^2 then the force on dipole is :-

(1) 0

(2) 2p

(3) p/2

(4) p

6) An electric dipole of length 2cm is placed with its axis making an angle of 30° to a uniform electric field 10^5 N/C. If it experiences a torque of $10\sqrt{3}$ Nm, then potential energy of the dipole is :-

(1) –10J

(2) –20J

(3) –30J

(4) –40J

7) Two wires have temperature coefficient α and –β and their resistance at 0°C temp are R_1 and R_2. Now these two wires are connected in series. If $\frac{R_1}{R_2} > \frac{\beta}{\alpha}$ then :-

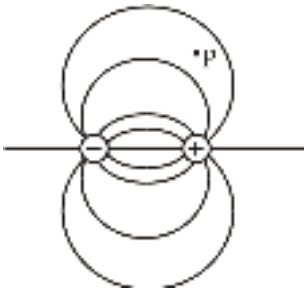
(1) Net resistance remains constant with respect to temperature

(2) Net resistance increases when temperature increases

(3) Net resistance increases when temperature decreases

(4) None

8) Figure shows the electric field lines around an electric dipole. Which of the arrows best



represents the electric field at point P ?

(1) 

(2) 

(3) 

(4) 

9) An α-particle of mass 6.4×10^{-27} kg and charge 3.2×10^{-19} C is situated in a uniform electric field of 1.6×10^{5} Vm^{-1}. The velocity of the particle at the end of 2×10^{-2} m path covered, when its starts from rest is :-

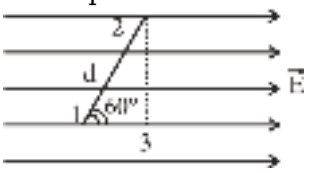
(1) $2\sqrt{3} \times 10^{5} ms^{-1}$

(2) 8×10^{5} ms^{-1}

(3) 16×10^{5} ms^{-1}

(4) $4\sqrt{2} \times 10^{5} ms^{-1}$

10) The electric field $\vec{E}$ between two points is constant in both magnitude and direction. Consider a path of length d at an angle θ = 60° with respect to field lines shown in figure. The potential



difference between points 1 and 2 is:-

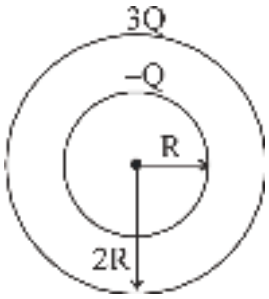
(1) $\frac{E}{d \cos 60^{\circ}}$

(2) Ed cos 60°

(3) $\frac{Ed}{\cos 60^{\circ}}$

(4) $\frac{E}{d} \cos 60^{\circ}$

11) Two conducting spherical shells of radii R and 2R carry charges – Q and 3Q respectively. How



much charge will flow into the earth if the inner shell is grounded ?

(1) $-\frac{Q}{2}$

(2) $\frac{Q}{2}$

(3) $\frac{3Q}{2}$

(4) $-\frac{3Q}{2}$

12) $V = x^2y - y^2 - x + 2y$. Find $\vec{E}$ at origin.

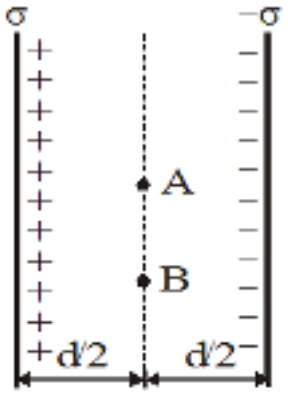
(1) $\left(\hat{i}+2\hat{j}\right)$ N/C

(2) $\left(-\hat{i}+2\hat{j}\right)$ N/C

(3) $\left(\hat{i}+2\hat{j}\right)$ N/C

(4) $\left(2\hat{i}-\hat{j}\right)$ N/C

13) Two particles A and B having charges '2q' and '–q' with masses 'm' and '2m' respectively are released from rest as shown then the ratio of the time taken by the two charge particle to strike the two non-conducting infinite sheets will be:-



(1) 1 : 2

(2) 2 : 3

(3) 1 : 4

(4) 3 : 5

14) Two negative charge, each of magnitude q are situated at 2r distance apart. A positive charge q is lying at the centre betweeen them. The potential energy of the system is U_1. If the two nearest

charges are mutually interchanged and the potential energy becomes U_2, then will be $\frac{U_1}{U_2}$:-

(1) 3

(2) 5

(3) 2

(4) 1

15) A uniformly charged non-conducting sphere of radius R has a potential V_0 at a distance 2R from the surface of sphere, then find the potential at a distance $\frac{R}{2}$ from center :-

(1) $\frac{11}{4}V_0$

(2) $\frac{33}{8}V_0$

(3) $\frac{33}{2}V_0$

(4) $\frac{11}{2}V_0$

16) A long wire of linear charge density λ is passed through the centre of spherical surface of radius R then flux through the surface is :-

(1) $\frac{\lambda R}{\epsilon_0}$

(2) $\frac{2\lambda R}{\epsilon_0}$

(3) $\frac{\sqrt{2}\lambda R}{\epsilon_0}$

(4) $\frac{\sqrt{3}\lambda R}{\epsilon_0}$

17) A tiny 0.50 gm ball carries a charge of magnitude 10 μC. It is suspended by a thread in a downward electric field of intensity 300 N/C. If the charge on the ball is positive, then the tension in the string is :

(1) 5×10^{-3} N

(2) 8×10^{-3} N

(3) 2×10^{-3} N

(4) zero

18) The capacitance of a parallel plate air capacitor is C_0. If a dielectric of dielectric constant K and thickness equal to one fourth the separation is placed between the plates, then capacity becomes C. Then ratio $\frac{C}{C_0}$ is-

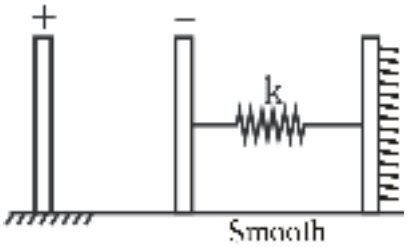
(1) $\frac{5K}{4K+1}$

(2) $\frac{4K}{3K+1}$

(3) $\frac{3K}{2K+1}$

(4) $\frac{2K}{K+2}$

19) Two charged capacitors have their outer plates fixed and inner plates connected by a spring of force constant 'k'. The charge on each capacitor is q and area of each plate is A. Find the extension in the spring at equilibrium



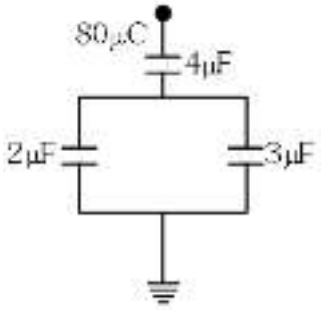
(1) $\frac{q^2}{2A\varepsilon_0 k}$

(2) $\frac{q^2}{4A\varepsilon_0 k}$

(3) $\frac{q^2}{A\varepsilon_0 k}$

(4) Zero

20) In the given circuit a charge of +80μC is given to the upper plate of 4μF capacitor. Then in the steady state, the charge on upper plate of the 3μF capacitor is :



(1) +32μC

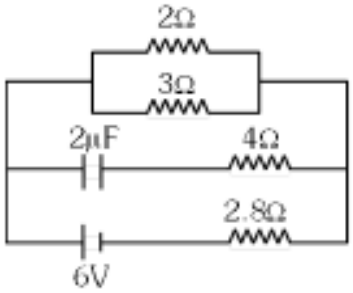
(2) +40μC

(3) +48μC

(4) +80μC

21)

In the figure shown, the capacitor of the capacity C is 2μF. The current in 2Ω resistor is :



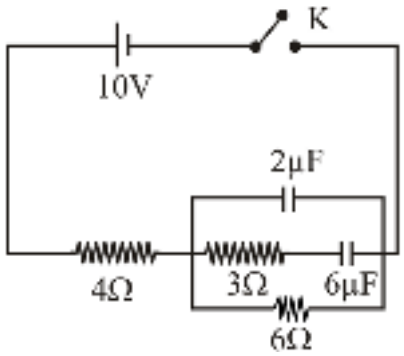
(1) 9A

(2) 0.9A

(3) $\frac{1}{9}$A

(4) $\frac{1}{0.9}$A

22) Determine the current through the battery in the circuit shown, Immediately after the key K is closed.



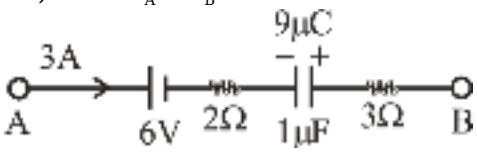
(1) 1.5 A

(2) 2A

(3) 1.7A

(4) 2.5A

23) Find $V_A - V_B$:-



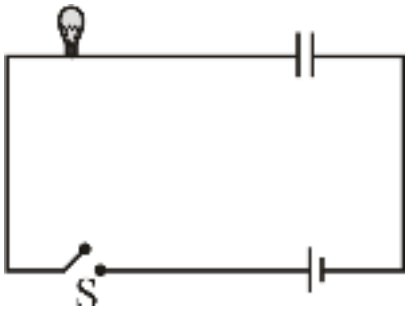
(1) 12V

(2) –12V

(3) 6V

(4) –6V

24) A light bulb, a capacitor and a battery are connected together as shown here, with switch S initially open. When the switch S is closed, which one of the following is true :-



(1) The bulb will light up for an instant when the capacitor starts charging.

(2) The bulb will light up when the capacitor is fully charged.

(3) The bulb will not light up at all

(4) The bulb will light up and go off at regular intervals.

25) Eight equally charged identical drops combine to form a big drop. If potential of each small drop is 10V, then potential of bigger drop is :

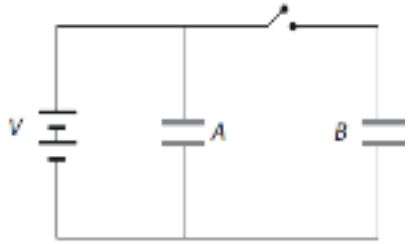
(1) 40 V

(2) 10 V

(3) 30 V

(4) 20 V

26) Figure given below shows two identical parallel plate capacitors connected to a battery with switch S closed. The switch is now opened and the free space between the plate of capacitors is filled with a dielectric of dielectric constant 3. What will be the ratio of total electrostatic energy stored in both capacitors before and after the introduction of the dielectric :-



(1) 3 : 1

(2) 5 : 1

(3) 3 : 5

(4) 5 : 3

27) In the diagram an isolated capacitor C is having charge Q and voltage V.
Now a material of dielectric constant K = 2 is inserted in the space between the plates as shown.
After inserting the slab



	Column I		Column II
a.	Charge on capacitor	p.	Becomes half of its initial value
b.	Potential difference across capacitor	q.	Remain same
c.	Energy stored in capacitor	r.	Becomes double its initial value
d.	Capacitance of capacitor		

Match the columns and choose the correct option

(1) a–q, b–p, c–p, d–r

(2) a–q, b–q, c–p, d–r

(3) a–q, b–p, c–r, d–r

(4) a–p, b–q, c–q, d–r

28) If the plates of a parallel plate capacitor connected to a battery are moved apart from each other, then
(A) the charge stored in it, increases.
(B) the energy stored in it, decreases.
(C) its capacitance decreases.
(D) the ratio of charge to its potential remains the same.
(E) the product of charge and voltage decreases.
Choose the most appropriate answer from the options given below :

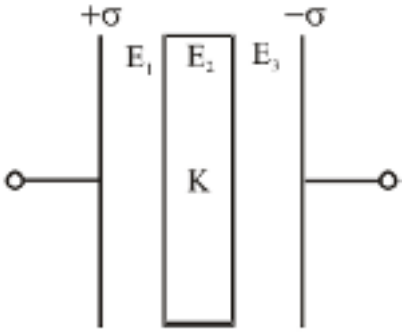
(1) B,C and E only

(2) A,C and E only

(3) B, D and E only

(4) A, B and C only

29) A dielectric slab of dielectric constant K is placed between the plates then compare electric fields E_1, E_2 and E_3.



(1) $E_1 = E_2 = E_3$

(2) $E_1 = E_3 > E_2$

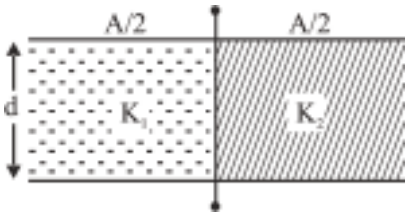
(3) $E_1 < E_2 < E_3$

(4) $E_1 = E_3 < E_2$

30)

A parallel plate condenser is filled with two dielectrics as shown in figure. Area of each plate is A

metre^2 and the separation is d metre. The dielectric constants are K_1 and K_2 respectively. Its capacitance in farad will be :



(1) $\frac{\varepsilon_0 A}{d}(K_1 + K_2)$

(2) $\frac{\varepsilon_0 A}{d}\left(\frac{K_1 + K_2}{2}\right)$

(3) $\frac{\varepsilon_0 A}{d}2(K_1 + K_2)$

(4) $\frac{\varepsilon_0 A}{d}\left(\frac{K_1 - K_2}{2}\right)$

31) Two insulated conducting charged spheres placed far away of radii R_1 and R_2 having charges Q_1 and Q_2 respectively are connected to each other, then there is :-

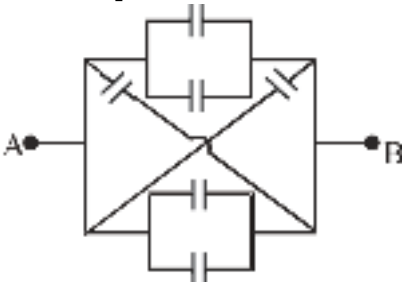
(1) No change in the energy of the system

(2) An increase in the energy of the system

(3) Always a decrease in the energy of the system

(4) A decrease in energy of the system unless $Q_1R_2 = Q_2R_1$

32) Six capacitors each of capacitance of 2μF are connected as shown in the figure. The effective capacitance between A and B is :-



(1) 12 μF

(2) $\frac{8}{3}$ μF

(3) 3 μF

(4) 6 μF

33)

A parallel plate capacitor having a plate separation of 2 mm is charged by connecting it to a 300 V supply. The energy density is :-

(1) 10 J/m^3

(2) 0.1 J/m^3

(3) 100 J/m^3

(4) 10^4 J/m^3

34) The resistance of an ammeter is 13 Ω and its scale is graduated for currents upto 100 A. After an additional shunt is connected to this ammeter it becomes possible to measure currents upto 750 amperes by this meter. The value of shunt-resistance is :–

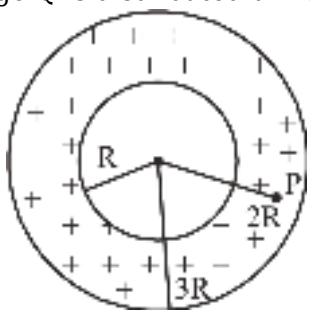
(1) 2 kΩ

(2) 20 Ω

(3) 2 Ω

(4) 0.2 Ω

35) A charge Q is distributed uniformly inside the shown hollow sphere. The electric field at point P is given by



(1) $E = \frac{7Q}{416\pi\in_0 R^2}$

(2) $E = \frac{9Q}{448\pi\in_0 R^2}$

(3) $\frac{7Q}{27\pi\in_0 R^2}$

(4) $\frac{7Q}{26\pi\in_0 R^2}$

36) A 1 Ω voltmeter has a range of 1V. Find the additional resistance which has to be joined with the series in voltmeter to increase the range of voltmeter to 100 V :–

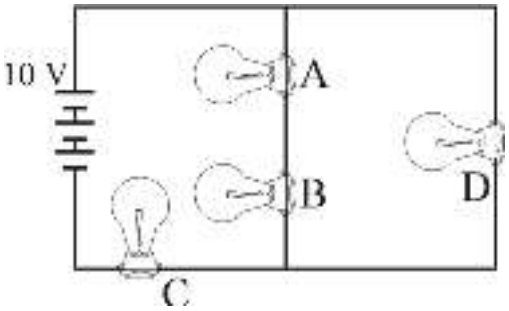
(1) 10 Ω

(2) $\frac{1}{99}$ Ω

(3) 99 Ω

(4) 100 Ω

37) In the circuit diagram, all the bulbs are identical. Which bulb will be the brightest?



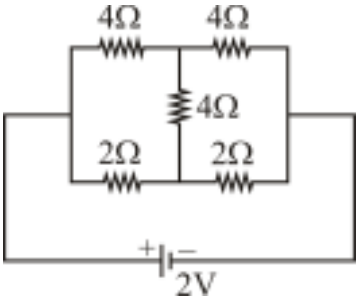
(1) A

(2) B

(3) C

(4) D

38) Find the power dissipated in the circuit.



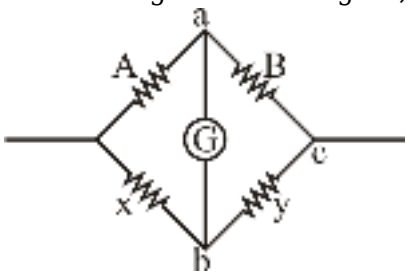
(1) 1.5 W

(2) 2 W

(3) 1 W

(4) none of the above

39) In a Wheatstone bridge shown in figure, x = y and A > B. The direction of the current between a and b will be



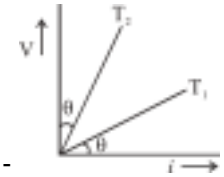
(1) from a to b

(2) from b to a

(3) from a to b through c

(4) from b to a through c

40) The V-i graph for a conductor at temperatures T_1 and T_2 are as shown in the figure (T_2–T_1) is proportional to :-



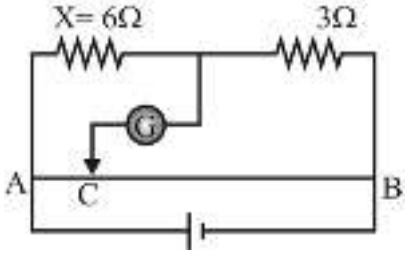
(1) cos 2θ

(2) sin θ

(3) cot 2θ

(4) tan θ

41) In given meter bridge circuit length of wire AB is 1m. When resistance X is shunted by R resistance then balancing length from A is found to be 20 cm. Find value of R :-



(1) $\frac{6}{7}\Omega$

(2) $\frac{7}{6}\Omega$

(3) $\frac{1}{3}\Omega$

(4) $\frac{2}{3}\Omega$

42) 100 cells, each of e.m.f. 5V and internal resistance 1Ω, are to be arranged so as to drive maximum current in a 25Ω resistance. Each row is to contain equal number of cells. The number of rows should be: -

(1) 2

(2) 4

(3) 5

(4) 10

43) The variation of current (I) and voltage (V) is as shown in figure A. The variation of power P with current I is best shown by which of the following graph

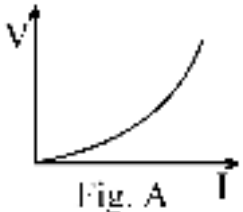
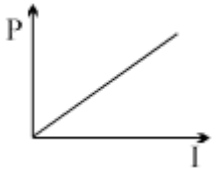


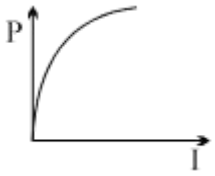
Fig. A

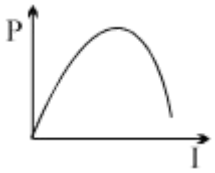
(1)



(2)



(3) 

(4) 

44) The area of cross section of current carrying conductor is A^0 and $\frac{A_0}{4}$ in section (1) & (2) if v_{d1}, v_{d2} and E_1, E_2 are drift velocity and electric field at sec. (1) and (2) respectively, then :-



(1) $v_{d1} : v_{d2} = 1 : 2$

(2) $v_{d1} : v_{d2} = 4 : 1$

(3) $E_1 : E_2 = 4 : 1$

(4) $E_1 : E_2 = 1 : 4$

45) Two wires A and B of equal masses and of the same metal are taken. The diameter of the wire A is half the diameter of the wire B. If the resistance of the wire A be 24 ohm, the resistance of the wire B is :-

(1) 3Ω

(2) 1.5 Ω

(3) 4.5 Ω

(4) 6Ω

CHEMISTRY

1) The secondary valency and the number of hydrogen bonded water molecule(s) in $CuSO_4.5H_2O$, respectively, are

(1) 7, 8

(2) 4, 1

(3) 6, 5

(4) 5, 1

2) Which of the following facts about the complex $[Cr(NH_3)_6]Cl_3$ is wrong ?

(1) The complex involves d^2sp^3 hybridisation

(2) The complex is paramagnetic.

(3) The complex is an outer orbital complex.

(4) The complex gives white precipitate with $AgNO_3$ solution.

3) In the cobalt-carbonyl complex: $[Co_2(CO)_8]$, number of Co-Co bonds is "X" and terminal CO ligands is "Y". X + Y =____

(1) 3

(2) 7

(3) 8

(4) 9

4) For the reaction given below :
$CoCl_3 . xNH_3 + AgNO_3 (aq) \rightarrow$
If two equivalents of AgCl precipitate out, then the value of x will be _____.

(1) 7

(2) 5

(3) 1

(4) 3

5) $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$ differ in :

(1) geometry, magnetic moment

(2) geometry, hybridization

(3) magnetic moment, colour

(4) hybridization, number of d-electrons

6) Which of the following is correctly matched ?

	Column-I	**Column-II**	**Column-III**
(A)	$[Cr(CO)_6]$	Paramagnetic	Octahedral, sp^3d^2
(B)	$[Fe(CO)_5]$	Paramagnetic	Trigonal bipyramidal, sp^3d
(C)	$K_4[FeCl_6]$	Paramagnetic	Octahedral, sp^3d^2
(D)	$[Ni(CO)_4]$	Diamagnetic	Square planar, dsp^2

(1) 1

(2) 2

(3) 3

(4) 4

7) Co-ordination compounds are used as catalyst for many industrial processes. Example include Rhodium complex used for the hydrogenation of alkenes, this catalyst is called as Wilkinson catalyst. What is true about Wilkinson catalyst ?

(1) Co-ordination number of Rhodium is three

(2) Oxidation number of Rhodium is +2

(3) Co-ordination number of Rhodium is six

(4) Oxidation number of Rhodium is +1

8) Which of the following complex is inner orbital as well as diamagnetic in nature ?

(1) $[Cr(NH_3)_6]^{3+}$

(2) MnO_4^{2-}

(3) $[NiCl_4]^{-2}$

(4) $[Co(H_2O)_6]^{3+}$

9) The degeneracy of the d-orbital is lost under the influence of :

(1) strong field ligand

(2) weak field ligand

(3) both (A) and (B)

(4) neither (A) nor (B)

10) Count tetrahedral complex compounds
$[Ni(CN)_4]^{-2}$,$[NiCl_4]^{-2}$,$[PtCl_4]^{-2}$, $[Ni(NH_3)_6]^{+2}$
$[Fe(CO)_5]$, $[Ni(CO)_4]$,$[Ag(NH_3)_2]Cl_2$

(1) 7

(2) 5

(3) 2

(4) 4

11) Count Number of complex which gives test of chloride ion
$[CoCl_6]^{-3}$,$[Pt(NH_3)_4Cl_2]Cl_2$,$[PtCl_4]^{-2}$
$[Ni(NH_3)_6]Cl_2$,$[Cr(OX)_3]^{-3}$,$[Pt(H_2O)_2Br_2Cl_2]$

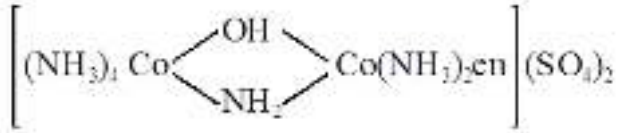
(1) 5

(2) 6

(3) 4

(4) 2

12) Oxidation state of cobalt in given complex –



are -

(1) +4

(2) +3

(3) +2

(4) +6

13) Potasium dicyanobis (oxalato) nickelate(II) formula is :-

(1) $K_3[Ni(CN)(OX)_2]$

(2) $K_3[Ni(CN)_2(OX)_2]$

(3) $K_4[Ni(CN)_2(OX)_2]$

(4) $K_2[Ni(CN)_2(OX)_2]$

14) Consider the complex ion, $[Cr(NH_3)_2Cl_2(C_2O_4)]^-$ .What is the :-
(i) Oxidation number of the metal ion ?
(ii) Co-ordination number of the metal ion ?
(iii) Charge on the complex if all ligands were chloride ions ?

	(i)	(ii)	(iii)
(A)	–1	6	–3
(B)	+3	6	–3
(C)	+3	6	–1
(D)	–1	5	–3

(1) A

(2) B

(3) C

(4) All are correct

15) Complexes $[Co(SO_4)(NH_3)_5]Br$ and $[CoBr(NH_3)_5]SO_4$ can be distinguished by : -

(1) Magnetic moment

(2) using $BaCl_2$

(3) using $AgNO_3$

(4) Both (2) and (3)

16) Which of the following complex have t_{2g}^4, e_g^2 configuration ?

(1) $[CoF_6]^{-3}$

(2) $[Co(H_2O)_6]^{+3}$

(3) $[Co(C_2O_4)_3]^{-3}$

(4) $[Co(CN)_6]^{-3}$

17) Wrong statement regarding $[Ni(CO)_4]$ is :-

(1) Sp^3 hybridised

(2) Unpaired electron is zero

(3) Diamagnetic complex

(4) Square planer geometry

18) In which of the following options all species contain X–O–X bonds in structure (X = central atom)

(1) $H_2S_2O_5$, S_3O_9, $S_2O_6^{-2}$

(2) P_4O_{10}, P_4O_6, $H_3P_3O_9$

(3) N_2O_5, N_2O, N_2O_4

(4) $H_4P_2O_7$, $H_4P_2O_6$, $H_4P_2O_5$

19) In how many of the following pairs, the first molecule has greater bond angle than the second molecule ?
(a) NH_3, NF_3 (b) PH_3, PF_3 (c) AsH_3, AsF_3
(d) AsF_3, $AsCl_3$ (e) H_2O, F_2O

(1) 3

(2) 2

(3) 4

(4) 5

20) Which of the following statement is correct ?

(1) SF_6 does not react with water

(2) OF_6 is d^2sp^3 hybridised

(3) In $S_2O_6^{2-}$ S–S bond is not present

(4) There is no π-bonding in SO_4^{2-}

21) Which of the following molecules is not a electron pair donor ?

(1) $(C_2H_5)NH$

(2) $(C_2H_5)_3N$

(3) $(SiH_3)_3N$

(4) NH_3

22) The decreasing order, Xe > Kr > Ar > Ne > He, is not correct for :-

(1) Boiling point

(2) Solubility in water

(3) Ionisation enthalpy

(4) Ease of liquefication

23) Which of the following is not true ?

(1) Among halide ions, iodide ion is the most powerful reducing agent

(2) Fluorine is the only halogen which does not show a variable oxidation state

(3) HOCl is stronger acid than HOBr

(4) HCl is stronger acid than HBr

24) In SiO_2 one Si attached with x oxygen atoms and one oxygen attached with y silicon atoms find (x+y)

(1) 3
(2) 4
(3) 6
(4) 7

25) Correct stability order of monohalide of Boron family :-

(1) BX > AlX > GaX > InX > TlX
(2) BX < AlX < GaX < InX < TlX
(3) BX > AlX < GaX > InX > TlX
(4) AlX > GaX > BX > InX > TlX

26) Which of the following gives propyne on hydrolysis

(1) Al_4C_3
(2) Mg_2C_3
(3) B_4C
(4) CaC_2

27)

Boric acid is polymeric due to

(1) Its acidic nature
(2) The presence of hydrogen bonds
(3) Its monobasic nature
(4) Its geometry

28) Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite :

(1) Has molecules of variable molecular masses like polymers
(2) Has carbon atoms arranged in large plated of rings of strongly bonded carbon atoms with weak interplate bonds
(3) Is a non crystalline substance
(4) Is an allotropic form of diamond

29) Number of peroxide linkage in CrO_5, H_2SO_5, $H_2S_2O_8$ are n_1, n_2 and n_3 respectively. Then find out the value of $(n_1+n_2+n_3)$

(1) 1
(2) 2
(3) 3

(4) 4

30) Which of the following reaction is not feasible

(1) $F_2 + 2Cl^- \longrightarrow 2F^- + Cl_2$

(2) $Cl_2 + 2Br^- \longrightarrow 2Cl^- + Br_2$

(3) $Br_2 + 2F^- \longrightarrow 2Br^- + F_2$

(4) $Br_2 + 2I^- \longrightarrow 2Br^- + I_2$

31) Which shows maximum bond energy :-

(1) Cl_2

(2) F_2

(3) Br_2

(4) I_2

32) Increasing order of electron gain enthalpy is :-

(1) Si > C > P > N

(2) O > S > Se

(3) F > Cl > Br

(4) All are correct

33) Match List I with List II and select the correct answer using the codes given below the lists :

List I		**List II**	
A.	BBr_3	i.	Dimer
B.	Tl_2O	ii.	Trigonal planar
C.	$B(OH)_3$	iii.	Basic
D.	$AlCl_3$	iv.	Monobasic acid

	A	**B**	**C**	**D**
(A)	i	ii	iii	iv
(B)	ii	iii	iv	i
(C)	iv	iii	i	ii
(D)	iii	iv	ii	iii

(1) A

(2) B

(3) C

(4) D

34) The thermal stability order for gp. 14 halides is :

(1) $CX_4 > SiX_4 > GeX_4 > SnX_4$

(2) $SnX_4 > GeX_4 > SiX_4 > CX_4$

(3) $SiX_4 > CX_4 > GeX_4 > SnX_4$

(4) None of these

35) Carbon dioxide is a gaseous molecule while SiO_2 is a three dimensional-network solid because :

(1) CO_2 molecules are held by strong vander Waal's forces whereas SiO_2 is an ionic solid

(2) CO_2 exists as a discrete molecule due to the formation of double bonds between the carbon and oxygen atoms by 2p-2p π-bonding while silicon cannot form such double bonds with oxygen because the 3pπ–2pπ overlap is not efficient; thus SiO_2 forms an infinite three-dimensional structure.

(3) CO_2 molecules are held by 2p–3p π - bonding and SiO_2 molecules are held by 3p-3d π overlap, thus SiO_2 forms an infinite three-dimensional structure

(4) None of these

36) Coinage metals show the properties of

(1) typical elements

(2) normal elements

(3) inner-transition elements

(4) transition element

37) Which of the following clement has maximum density ?

(1) Hg

(2) Au

(3) Os

(4) Pb

38) Which of the following has highest atomic radii ?

(1) Li

(2) Mn

(3) Fe

(4) Co

39) Which of the following set of elements have almost same radii :-

(1) Y & La

(2) Ti & Zr

(3) Mo & W

(4) V & Nb

40) Which of the following compound does not exist :-

(1) VF_5

(2) V_2O_5

(3) VI_5

(4) Cr_2O_3

41) In which of the following oxo-anion, all M–O bond length are not identical ?

(1) MnO_4^-

(2) MnO_4^{-2}

(3) $Cr_2O_7^{-2}$

(4) CrO_4^{-2}

42) K_2MnO_4 can be converted into $KMnO_4$ by :-

(1) Passing CO_2 gas

(2) By passing Cl_2

(3) Electrolytic oxidation

(4) All of these

43)

Brass is an alloy of :-

(1) Silver and copper

(2) Copper and zinc

(3) Copper and tin

(4) Copper, zinc and tin

44) The ionic radius of Cr is minimum in which of the following compounds ?

(1) CrF_3

(2) $CrCl_3$

(3) Cr_2O_3

(4) K_2CrO_4

45) Which form interstitial compounds ?

(1) Fe

(2) Ni

(3) V

(4) All of these

BIOLOGY

1) Other membranous extensions into the cytoplasm in blue green algae are known as :

(1) phytochrome
(2) chromatophore
(3) Inclusion boies
(4) pneumatophore

2) An organelle devoid of membrane covering is -

(1) vacuole
(2) ribosome
(3) peroxisome
(4) lysosome

3) The membrane of the erythrocytes has approximately ___% of proteins and ___% lipids.

(1) 42, 50
(2) 52, 40
(3) 50, 50
(4) 60, 40

4) The best material for study of structure of cell membrane is -

(1) RBC of human
(2) RBC of frog
(3) Cheek cell of human
(4) Liver cell of rat

5) Which one of the following structures between two adjacent cells is an effective transport pathway?

(1) Plasmodesmata
(2) Plastoquinones
(3) Endoplasmic reticulum
(4) Plasmalemma

6) Plastids storing fat are called :

(1) Elaioplasts
(2) Sphaerosomes
(3) Aleuroplasts
(4) Pyrenoids

7) Axoneme with 9 + 2 microtubular arrangement occurs in

(1) Cilia
(2) Flagella
(3) Both (1) and (2)

(4) Centriole

8) Which of the following statement of a bacterial cell is/ are correct?
(i) Mesosome is formed by the extensions of plasma membrane into the cell.
(ii) The pili are elongated tubular structures made up of a protein.
(iii) Flagellum is composed of filament, hook and basal body.
(iv) Ribosomes are about 30 nm by 50 nm in size.

(1) (i), (ii) and (iii)

(2) All of the above

(3) (ii) and (iv)

(4) None of the above

9) Study the following statements on cilium or flagellum and answer the question.
(i) Cilium / Flagellum contains an outer ring of nine doublet microtubules surrounding two singlet microtubules.
(ii) Cilia are smaller which work like oars, causing the movement of either the cells or surrounding fluid.
(iii) Flagella are comparatively longer and responsible for cell movement.
(iv) Cilium and flagellum are covered with plasma membrane.
Which of the above statements are correct?

(1) (i) and (ii)

(2) (i), (ii), (iii) and (iv)

(3) (i) and (iv)

(4) (ii) and (iii)

10) Choose the wrong statements regarding bacterial cell :
(i) Glycocalyx is the outermost envelope in bacteria.
(ii) The glycocalyx could be a loose sheath called capsule.
(iii) The glycocalyx may be thick and tough called slime layer.
(iv) A special structure formed by the plasma membrane is called mesosome.
(v) Small bristle like fibres sprouting out of the cell are called fimbriae.

(1) (i) and (iii)

(2) (i) and (ii)

(3) (ii) and (iii)

(4) (i) and (iv)

11) Which of the following statements is/are correct ?
(i) The endomembrane system includes plasma membrane, ER, Golgi complex, lysosomes and vacuoles.
(ii) ER helps in the transport of substances, synthesis of proteins and lipoproteins.
(iii) Ribosomes are involved in protein synthesis.
(iv) Mitochondria help in oxidative phosphorylation and generation of ATP.

(1) (ii), (iii) & (iv)

(2) (i) only

(3) (ii) only

(4) (iii) only

12)

Read the statements given below with regard to the functions performed by Golgi apparatus?
(i) Transport and chemically modify the materials contained within it.
(ii) Performs the function of packaging materials.
(iii) Important site of formation of glycoproteins and glycolipids.
Which of the following is the correct answer?

(1) (i) is wrong but (ii) and (iii) are correct

(2) (ii) is wrong but (i) and (iii) are correct

(3) (ii) and (iii) are wrong but (i) is correct

(4) All are correct

13) Which of the following statements are correct?

(1) Na^+/K^+ pump is an example of active transport.

(2) In plant cells lipid like steroidal hormones are synthesized in RER.

(3) In plant cells, the vacuoles can occupy up to 10% of the volume of the cell.

(4) Chlorophyll and leucoplast are responsible for trapping light energy essential for photosynthesis.

14) **Assertion :** Centrosomes and centrioles are related to each other.
Reason : Centrosome usually contains two cylindrical structures called centrioles.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

15) **Assertion :** The acrocentric chromosome has centromere at the terminal position.
Reason : The metacentric chromosome has centromere slightly away from the middle of the chromosome.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

16) Match column-I (scientists) with column-II (discovery) and select the correct option.

Column-I		**Column-II**	
A	Leeuwenhoek	I	First saw and described a living cell

B	Robert Brown	II	Presence of cell wall is unique to plant cells
C	Schleiden	III	Discovered the nucleus
D	Schwann	IV	All plants are composed of different kind of cells

(1) A – I; B – III; C – IV; D – II

(2) A – I; B – III; C – II; D – IV

(3) A – III; B – I; C – IV; D – II

(4) A – I; B – IV; C – II; D – III

17) Match column I (cell type) with column II (size) and choose the correct option.

Column-I (Cell type)		**Column-II (Size)**	
A.	Viruses	I.	1-2 µm
B.	PPLO	II.	10-20 µm
C.	Eukaryotic cell	III.	About 0.1 µm
D.	Bacterium	IV.	0.02 - 0.2 µm

(1) A – I; B – II; C – III; D – IV

(2) A – IV; B – III; C – II; D – I

(3) A – I; B – III; C – II; D – IV

(4) A – IV; B – II; C – III; D – I

18) Match column-I with column-II and choose the correct option.

Column-I		**Column-II**	
A	Tonoplast	I	Contain digestive enzyme
B	Contractile vacuole	II	Store metabolic gases
C	Food vacuole	III	Excretion
D	Air vacuole	IV	Transport of ions in plants

(1) A – IV; B – III; C – I; D – II

(2) A – II; B – III; C – IV; D – I

(3) A – IV; B – II; C – III; D – I

(4) A – I; B – III; C – II; D – IV

19) Which one of the following pairs is NOT correctly matched?

(1) Cristae – The tubular structure formed by the folding of the inner membrane of the mitochondrion.

(2) Plasmodesmata – The membrane surrounding the vacuole in plants.

(3) Grana – Membrane bound discs in chloroplasts that contain chlorophylls and carotenoids.

(4) Middle lamella – Layer between adjacent cells walls in plants derived from cell plate.

20) Which one of the following cellular parts is correctly matched with their characters ?

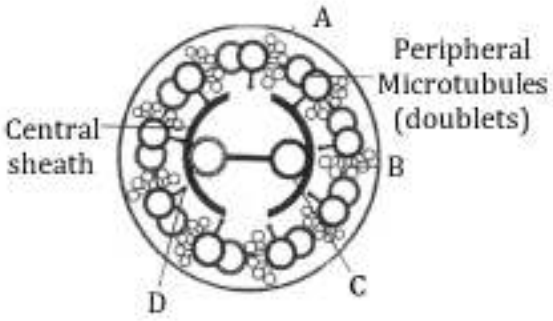
(1) Centrioles – Sites for active RNA synthesis.

(2) Lysosomes – Optimally active at a pH of about 8.5.

(3) Thylakoids – Flattened membranous sacs forming the grana of chloroplasts.

(4) Ribosomes – Those on chloroplasts are larger (80S) while those in the cytoplasm are smaller (70S).

21) Identify the components labelled A, B, C and D in the given section of cilia/flagella showing different parts. Choose the option which shows the correct labelling of parts.



(1) A – Plasma membrane, B – Inter doublet bridge, C – Central microtubule, D – Radial spoke

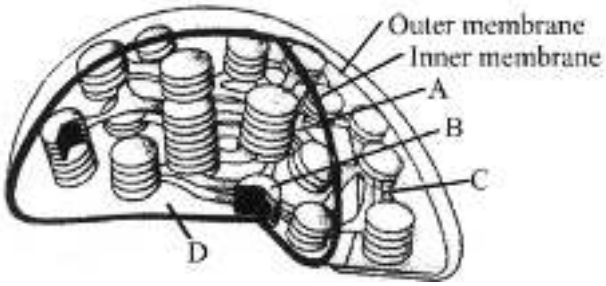
(2) A – Plasma membrane, B – Arm, C – Central microtubule, D – Radial spoke

(3) A – Plasma membrane, B – Inter doublet bridge, C – Hub, D – Radial spoke

(4) A – Plasma membrane, B – Inter doublet bridge, C– Hub, D – Arm

22)

Which of the following option correctly match A, B, C, and D indicated in the given sectional view of chloroplasts.



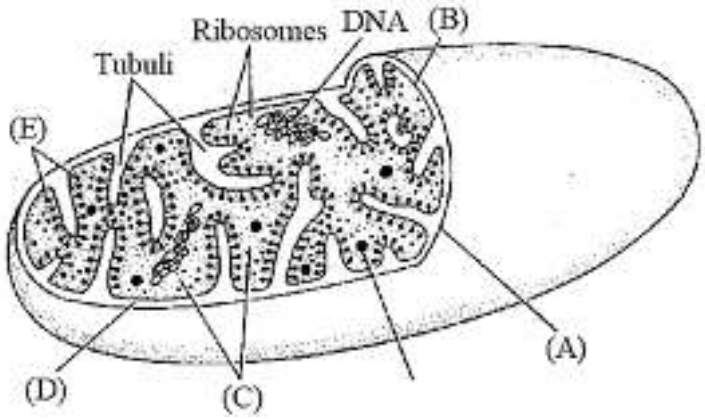
(1) A - Thylakoid, B-Stromal lamella, C - Stroma, D - Granum

(2) A - Granum, B - Thylakoid, C - Stromal lamella, D - Stroma

(3) A - Thylakoid, B - Granum, C - Stromal lamella, D - Stroma

(4) A - Granum, B - Thylakoid, C - Stroma, D – Stromal lamella

23) The given diagram shows the sectional view of a mitochondrion.



Identify the parts labelled as A, B, C, D & E

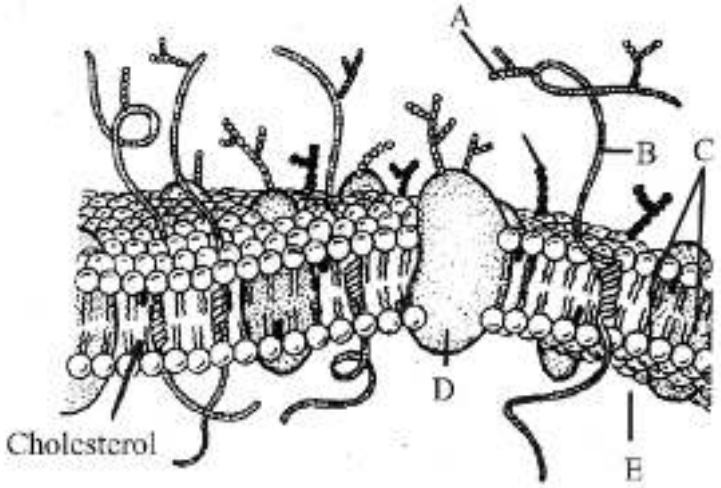
(1) A - Outer membrane, B - Inner membrane, C - Matrix, D - Inter- membrane space, E - Crista

(2) A - Outer membrane, B - Inner membrane, C – Intermembrane space, D - Matrix, E - Crista

(3) A - Outer membrane, B - Inner membrane, C - Matrix, D - Crista, E - Inter - membrane space

(4) A - Outer membrane, B - Inner membrane, C- Crista, D - Matrix, E - Inter-membrane space

24) The diagram given below represent a fluid mosaic model of plasma membrane. Match the components marked as A, B, C, D and E in the diagram below from the list (i) to (vii).



(i) Sugar (ii) Protein (iii) Lipid bilayer (iv) Integral protein (v) Cytoplasm (vi) Cell wall (vii) External protein

(1) A - (i), B - (ii), C - (iii), D - (iv), E - (v)

(2) A - (ii), B - (i), C - (iii), D - (iv), E - (v)

(3) A - (i), B - (ii), C - (iii), D - (iv), E - (vi)

(4) A - (i), B - (ii), C - (iii), D - (vii), E - (v)

25) The following diagram represents a structure chromosome. Identify the structures marked as A,

B and C. 

(1) A → Satellite, B → Primary constriction, C → Acrocentric
(2) A → Satellite, B → Secondary constriction, C → Metacentric
(3) A → Satellite, B → Centromere, C → Telocentric
(4) A → Satellite, B → Centromere, C → Submetacentric

26) What would happen if lysosomes get ruptured in a cell?

(1) Cell dies
(2) Cell shrinks
(3) Cell swell up
(4) Nothing would happen

27) In which method of transport, plasma membrane does not require carrier molecule?

(1) Active transport
(2) Facilitated diffusion
(3) Simple diffusion
(4) Na^+ – K^+ pump

28) pH of vacuolar cell sap is -

(1) neutral and isotonic
(2) alkaline and isotonic
(3) acidic and hypertonic
(4) equal to cytoplasm and isotonic

29) Which of the following pair lack the unit membrane?

(1) Nucleus and E.R.
(2) Mitochondria and chloroplast
(3) Ribosome and nucleolus
(4) Golgi body and lysosome

30) You are asked to examine a cell using a powerful light microscope. The image you see has a clearly defined nucleus and mitochondria. It also has a large central vacuole and chloroplasts. From what group of organisms did this cell most likely come?

(1) Bacteria
(2) Protists
(3) Fungi
(4) Plants

31) Which phase of cell cycle is known as quiescent stage ?

(1) M phase
(2) G_0 phase
(3) G_1 phase
(4) S phase

32) In mitosis, nucleolus and nuclear membrane disappear at :

(1) interphase
(2) prophase
(3) metaphase
(4) telophase

33) In meiosis, division is -

(1) I^{st} reductional and IInd equational
(2) I^{st} equational and IInd reductional
(3) both reductional
(4) both equational

34) Synapsis occurs between :

(1) spindle fibres and centromeres
(2) mRNA and ribosomes
(3) a male and female gamete
(4) two homologous chromosomes

35) The synaptonemal complex appears :

(1) between homologous chromosomes.
(2) in zygotene stage.
(3) composed of DNA + protein.
(4) All of the above

36) In which stage, the chromosomes appear thin and long thread-like ?

(1) Zygotene
(2) Leptotene
(3) Pachytene

(4) Prophase

37) Which of the following statement is not true for homologous chromosome pairs ?

(1) They come from only one of the individual's parents.
(2) They usually contain slightly different versions of the same genetic information.
(3) They segregate from each other during meiosis I.
(4) They synapse during meiosis I.

38) Cell plate grows from -

(1) wall to centre
(2) centre to walls
(3) one wall to another
(4) simultaneously

39) Terminalization occurs during _____

(1) mitosis
(2) diakinesis
(3) meiosis II
(4) cytokinesis

40) Which of the following statement(s) is/are true ?

(1) Cell plate represents the middle lamella between the walls of two adjacent plant cells.
(2) At the time of cytokinesis, organelles like mitochondria and plastids get distributed between the daughter cells.
(3) Cytokinesis in plant cell is centrifugal and takes place by cell-plate formation while animal cells by furrowing/cleavage and is centripetal.
(4) All of the above

41) Which of the following statement(s) is/are correct about S-phase (synthetic phase)?
(i) It occurs between G_1 and G_2 phase.
(ii) It marks the period during which DNA replicates.
(iii) At the end of this phase, DNA is doubled but the number of chromosomes remains unchanged.
(iv) As the DNA is doubled in this phase number of chromosomes is also doubled.
(v) Centrioles replicate in this phase.
(vi) Amount of DNA changes from 2C to 4C.

(1) (i), (ii), (iv), (v), (vi)
(2) (i), (ii), (iii), (v), (vi)
(3) All of the above
(4) Only (iv)

42)

Which of the following statements related to G0 stage of the cell cycle are correct?
(i) It is a quiescent stage.
(ii) In this phase, cell cycle is suspended.
(iii) G_0 cells do not grow or proliferate but metabolically active.
(iv) G_0 cells can divide in response to some stimulus.

(1) (i), and (ii)

(2) (i), (ii) and (iii)

(3) Only (i) and (iv)

(4) All of the above

43) Which one is correct about crossing over/genetic recombination?
(i) It occurs in tetrad stage which occurs in pachytene of prophase I of meiosis I.
(ii) It occurs between non-sister chromatids of homologous chromosomes.
(iii) It is recombinase enzyme mediated process.
(iv) It is also the reciprocal transfer of genes between the non-homologous chromosomes.

(1) All of the above

(2) All except (iv)

(3) (ii), (iii) and (iv)

(4) (iii) and (iv)

44) **Assertion:** Diplotene is characterized by the presence of chiasmata.
Reason: Diplotene can last for months and years in oocytes of some vertebrates.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

45) **Assertion:** Small disc-shaped structures at the surface of the centromeres are called kinetochores.
Reason: Kinetochores serve as the sites of attachment of spindle fibres to the centrosome.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

46) **Assertion:** The crossing over is an enzyme-mediated process.
Reason: The enzyme involved in crossing over is lyase.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

47) **Assertion:** Metaphase II begins with splitting of centromere of each chromosome into two.
Reason: In Anaphase II chromosomes align at the equator.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

48) Match the description (given in column I) with correct stage of prophase I (given column II) and choose the correct option.

Column-I		**Column-II**	
A	Chromosomes are moved to spindle equator	I	Pachytene
B	Centromere splits and chromatids apart	II	Zygotene
C	Pairing between homologous chromosomes takes place	III	Anaphase
D	Crossing between homologous chromosomes	IV	Metaphase

(1) A – I; B – II; C – III; D – IV

(2) A – II; B – III; C – IV; D – I

(3) A – IV; B – III; C – II; D – I

(4) A – III; B – I; C – IV; D – II

49) Match the terms (given in column I) with their explanation (given in column II) and choose the correct combination from the options given below.

Column-I (Terms)		**Column-II (Explanation)**	
A.	Terminalization	I.	Pairing of homologous chromosomes.
B.	Synapsis	II.	X-shaped structures.
C.	Chiasmata	III.	Nuclear protein complex that helps in adherence of sister chromatids and then homologous chromosomes.
D.	Synaptonemal complex	IV.	Shifting of chiasmata Outwards complex towards the ends of a bivalent.

(1) A – IV; B – I; C – II; D – III

(2) A – II; B – III; C – IV; D – I

(3) A – II; B – IV; C – III; D – I

(4) A – IV; B – I; C – III; D – II

50) Match the description given in column-I with their stapes given in column-II and identify the correct answer.

Column-I		**Column-II**	
A	Initiation of the assembly of mitotic spindle	I	Anaphase

B	Proteins are synthesized in preparation for mitosis while cell growth continues	II	Prophase
C	Spindle fibres attach to kinetochores of chromosomes.	III	Interphase
D	Movement of chromatids towards opposite poles	IV	Metaphase

The correct match is -

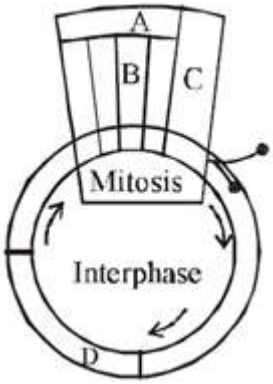
(1) A – II; B – III; C – IV; D – I

(2) A – III; B – II; C – I; D – IV

(3) A – I; B – III; C – II; D – IV

(4) A – IV; B – III; C – I; D – II

51) Given below is a schematic break-up of the phases / stages of cell cycle with few parts labelled as A, B, C & D.



Which one of the following is the correct indication of the stage/phase in the cell cycle?

(1) C - Karyokinesis

(2) D - Synthetic phase

(3) A - Cytokinesis

(4) B - Metaphase

52) Which stages of cell division do the following figures A and B represent respectively?

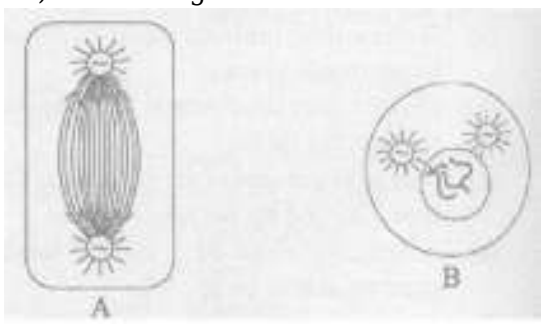


	Fig. A	Fig. B
(1)	A - Metaphase	B - Telophase
(2)	A - Telophase	B - Metaphase
(3)	A - Late Anaphase	B - Prophase
(4)	A - Prophase	B - Anaphase

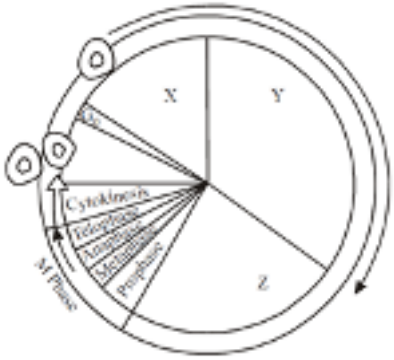
(1) 1

(2) 2

(3) 3

(4) 4

53) The following diagram is of a typical cell cycle indicating formation of two cells from one cell.



Choose the correct option showing the correct identification of X, Y, & Z.

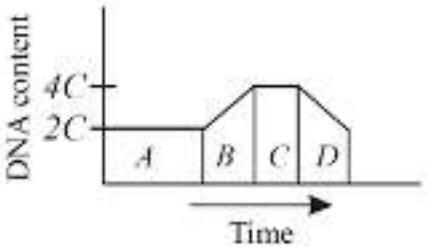
(1) X - G_1; Y - S; Z - G_2

(2) X - G_2; Y - S; Z - G_1

(3) X - G_0; Y - S; Z - G_2

(4) X - G_1; Y - G_0; Z - S

54) Identify the phases from the graph given below that shows the change in DNA content during various phases (A to D) of mitotic cell cycle.



	A	B	C	D
(1)	G_2	G_1	S	M
(2)	G_2	S	G_1	M
(3)	G_1	S	G_2	M
(4)	M	G_1	S	G_2

(1) 1

(2) 2

(3) 3

(4) 4

55) How many meiotic divisions are required to produce 100 pollen grains?

(1) 25

(2) 50

(3) 100

(4) 125

56) During which stages (or prophase I substages) of meiosis do you expect to find the bivalents and DNA replication respectively?

(1) Pachytene and interphase (between two meiotic divisions).

(2) Pachytene and interphase (just prior to prophase I).

(3) Pachytene and S phase (of interphase just prior to prophase I).

(4) Zygotene and S phase (of interphase prior to prophase I).

57) 'X' ensures the production of 'Y' phase in the life cycle of sexually reproducing organisms whereas fertilization restores the 'Z' phase.
Identify X, Y and Z.

(1) X- Mitosis, Y - haploid, Z – haploid

(2) X- Mitosis, Y - diploid, Z - diploid

(3) X- Meiosis, Y - haploid, Z – diploid

(4) X- Meiosis, Y - diploid, Z - diploid

58) The following events occur during the process of meiosis. Arrange the following events of meiosis in the correct sequence.
I. Terminalisation
II. Crossing over
III. Synapsis
IV. Disjunction of genomes

(1) I, II, III, IV

(2) II, III, IV, I

(3) III, II, I, IV

(4) IV, I, II, III

59) Four daughter cells formed after meiosis are -

(1) genetically similar

(2) genetically different

(3) anucleate

(4) multinucleate

60) In animal cells, cytokinesis involves -

(1) the separation of sister chromatids.

(2) the contraction of the contractile ring of microfilament.

(3) depolymerization of kinetochore microtubules.

(4) a protein kinase that phosphorylates other enzymes.

61) Which of the following is wrongly matched?

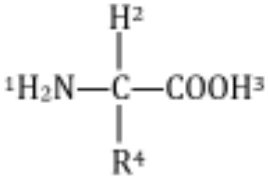
(1) Ribozyme – Proteinaceous in nature.

(2) Apoenzyme – The protein part of enzyme.

(3) Co-enzyme – Loosely attached organic cofactor of holoenzyme.

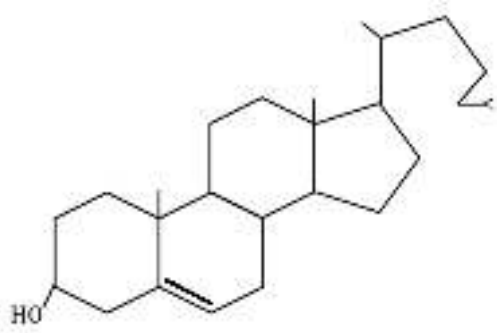
(4) Co-factors – Non-protein part of holoenzyme.

62) Which of the two groups (marked as 1, 2, 3, & 4) of following formula involved in peptide bond



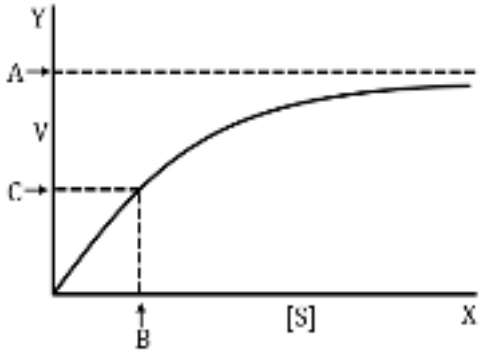
between different amino acids?

(1) 2 and 3
(2) 1 and 3
(3) 1 and 4
(4) 2 and 4

63) Identify the following molecule. 

(1) Phospholipid
(2) Lecithin
(3) Cholesterol
(4) Oleic acid

64) The adjoining graph shows change in concentration of substrate on enzyme activity.



Identify A, B and C.

	A	**B**	**C**
(1)	Ki	K_m	V_{max}
(2)	$\frac{V_{max}}{2}$	K_m	Ki
(3)	V_{max}	K_m	$\frac{V_{max}}{2}$
(4)	K_m	V_{max}	$\frac{V_{max}}{2}$

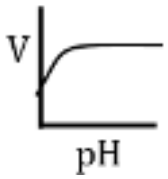
(1) 1

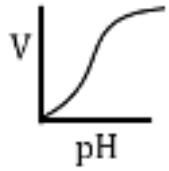
(2) 2

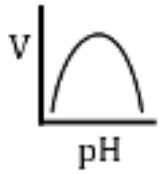
(3) 3

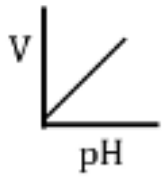
(4) 4

65) Which one of the given graph shows the effect of pH on the velocity of a typical enzymatic reaction (V) ?

(1) 

(2) 

(3) 

(4) 

66) Ligases is a class of enzymes responsible for catalysing the linking together of two compounds. Which of the following bonds is not catalysed by it?

(1) C – C

(2) P – O

(3) C – O

(4) C – N

67) Match column I (organic compound) with column II (examples) and choose the correct combination from the given options.

Column I (Organic Compounds)		**Column II (Example)**	
A.	Fatty acid	I.	Glutamic acid
B.	Phospholipid	II.	Tryptophan
C.	Aromatic amino acid	III.	Lecithin
D.	Acidic amino acid	IV.	Palmitic acid

(1) A – I; B – II; C – III, D – IV

(2) A – IV; B – III; C – II, D – I

(3) A – II; B – III; C – IV, D – I

(4) A – III; B – IV; C – I, D – II

68) Match the protein given in column I with its function given in column II and choose the right option.

Column I (Proteins)		**Column II (Functions)**	
A.	Collagen	I.	Glucose transport
B.	Trypsin	II.	Hormone
C.	Insulin	III.	Intercellular ground substance
D.	GLUT-4	IV.	Enzyme

(1) A – III; B – IV; C – II; D – I

(2) A – IV; B – I; C – II; D – III

(3) A – II; B – IV; C – I; D – III

(4) A – III; B – IV; C – I; D – II

69) Match column I (function) with column II (Types of enzymes) and select the correct option.

	Column I (Function)		**Column II (Types of enzymes)**
A.	Enzymes catalysing breakdown without addition of water	I.	Hydrolases
B.	Enzyme catalyzes the breakdown with addition of water.	II.	Oxidoreductase
C.	Enzyme where catalysis involves transfer of electrons.	III.	Ligases
D.	Enzyme catalysing bonding of two components with the help of ATP.	IV.	Lyases

(1) A – I; B – IV; C – III; D – II

(2) A – I; B – IV; C – II; D – III

(3) A – IV; B – I; C – II; D – III

(4) A – IV; B – I; C – III; D – II

70) Which one of the following pairs of nitrogenous bases of nucleic acids, is wrongly matched with the category mentioned against it ?

(1) Adenine, thymine – Purines

(2) Thymine, uracil – Pyrimidines

(3) Uracil, cytosine – Pyrimidines

(4) Guanine, adenine – Purines

71) Most abundant protein in human body is :

(1) Collagen

(2) Myosin

(3) Actin

(4) Albumin

72) **Assertion :** Unsaturated fats are more reactive compared with the saturated fats.

Reason : Unsaturated fats have only single bonds in their structure.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

73) Which of the following statement is not true about amino acid?

(1) The amino group and carboxylic group of an amino acid are attached to α carbon.

(2) Proteins have an N-terminus and an C-terminus.

(3) Amino acids can be classified by the number of peptide bond.

(4) Essential amino acids are not synthesized in the body, therefore have to be provided in the diet.

74) Select the incorrect statement.

(1) Ribozymes are nucleic acids with catalytic power.

(2) Proteins are homopolymer made of amino acids.

(3) Inulin is a polymer of fructose.

(4) Glycogen is stored in liver and muscles.

75) Which of the following statement is wrong regarding chitin?

(1) It is a storage of polysaccharide.

(2) It is a homopolysaccharide.

(3) It is a constituent of arthropods and fungal cell wall.

(4) It has N-acetyl glucosamine as monomer.

76) Which of the following statements about enzymes is incorrect?

(1) Enzymes are denatured at high temperature but in certain exceptional organisms. They are effective even at 80° - 90°C.

(2) Enzymes require optimum pH for maximal activity.

(3) Most enzymes are proteins but some are lipids.

(4) Enzymes are highly specific.

77) Select false statements for an enzyme promoting a chemical reaction by -
(i) lowering the energy of activation.
(ii) causing the release of heat, which acts as a primer.
(iii) increasing molecular motion.
(iv) changing the free energy difference between substrate and product.

(1) (i) and (iv)

(2) (ii) and (iii)

(3) (ii), (iii) and (iv)

(4) (iii) and (iv)

78) Inorganic catalyst work efficiently at _____temperature and _____ pressure.

(1) high, low
(2) low, low
(3) low, high
(4) high, high

79) Co-enzyme is :

(1) Always a protein
(2) Often a metal
(3) Always an inorganic compound
(4) Often a vitamin

80) The Km value of the enzyme is the value of the substrate concentration at which the reaction reaches to -

(1) Zero
(2) 2 V_{max}
(3) ½ V_{max}
(4) ¼ V_{max}

81) The simplest amino acid is _____.

(1) Glycine
(2) Proline
(3) Leucine
(4) Tryptophan

82) Unsaturated fatty acids have -

(1) Palmitic acid
(2) Only single bond
(3) High melting point
(4) One or more double bonds

83) The acid used for preliminary separation of biomolecule in a living tissue is :

(1) Trichlorobenzoic acid
(2) Benzoic acid
(3) Trichloroacetic acid
(4) Acetic acid

84) All the following amino acids are aromatic, except -

(1) Tyrosine

(2) Phenylalanine
(3) Tryptophan
(4) Valine

85) Phospholipids are important cell membrane constituents, because they :

(1) Contain glycerol
(2) Can form bilayers in water
(3) Combine covalently with protein
(4) Contain polar and non-polar portions

86) Lecithin, a small molecular weight organic compound found in living tissues, is an example of :

(1) Amino acids
(2) Phospholipids
(3) Glycerides
(4) Carbohydrates

87) Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of :

(1) Cofactor inhibition
(2) Feedback inhibition
(3) Competitive inhibition
(4) Enzyme activation

88) The cofactor of the enzyme carboxypeptidase is :

(1) Zinc
(2) Niacin
(3) Flavin
(4) Haem

89) Regarding catalytic cycle of an enzyme action, select the correct sequential steps :
A. Substrate enzyme complex formation.
B. Free enzyme ready to bind with another substrate.
C. Release of products.
D. Chemical bonds of the substrate broken.
E. Substrate binding to active site.
Choose the correct answer from the options given below:

(1) E, A, D, C, B
(2) A, E, B, D, C
(3) B, A, C, D, E
(4) E, D, C, B, A

90) Haem is an example of :

(1) Coenzyme
(2) Prosthetic group
(3) Co-factor
(4) Both (2) and (3)

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	2	2	4	3	2	2	4	2	2	3	1	1	2	2	2	2	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	4	1	1	1	3	1	1	2	2	4	1	2	3	1	3	3	1	2	3
Q.	41	42	43	44	45															
A.	1	1	2	4	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	2	2	3	3	4	4	3	3	4	2	3	2	4	1	4	2	2	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	4	3	2	2	2	2	4	3	1	1	2	1	2	4	3	2	3	3
Q.	86	87	88	89	90															
A.	3	4	2	4	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	2	2	1	1	1	3	1	2	3	1	4	1	1	4	1	2	1	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	2	1	1	2	1	3	3	3	4	2	2	1	4	4	2	1	2	2	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	2	2	3	3	4	3	1	1	2	3	1	3	1	4	3	3	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	2	3	3	3	1	2	1	3	1	1	3	3	2	1	3	3	4	4	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	4	3	4	4	2	3	1	1	4										

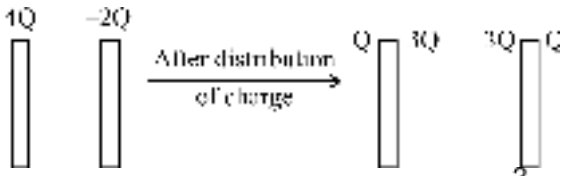
SOLUTIONS

PHYSICS

1) **Explaining :** find potential energy stored

Concept : This question is based on $u = \frac{q^2}{2C}$

Solution :



Energy store between two plates. $= \frac{q^2}{2C}$

where q = charge on capacitor.

 q = 3Q

So, energy store $= \frac{(3Q)^2}{2C} = \frac{9}{2}\frac{Q^2}{C}$

Final Answer : (2)

2) **Explaining :** A and B are concentric conducting spherical shells. A is given a positive charge while B is earthed. Then :-

Concept : This question is based on property of conductor

Solution :

(1) $\rho_A \neq \rho_B$

$\because \rho = \frac{Q}{V}$

(2) $V_A \neq 0$

$V_B \neq 0$

(3) $E \neq 0$

(4) $E_A = 0$

Final Answer : (3)

3)

Explaining : Find flux

Concept : This question is based on $\phi = \frac{q}{\varepsilon_0}$

Solution :

Step 1: Apply Gauss' law to get total flux: $\phi_{total} = \frac{q}{\epsilon_0}$

Step 2: Recognize symmetry implies flux through plane surfaces A and C are equal: $\phi_A = \phi_C$

Step 3: Express total flux in terms of known and unknown fluxes: $\frac{q}{\epsilon_0} = \phi + 2\phi_A$

$$2\phi_A = \frac{q}{\epsilon_0} - \phi \implies \phi_A = \frac{1}{2}\left(\frac{q}{\epsilon_0} - \phi\right)$$

Step 4: Solve for ϕ_A:

Final Answer : (2)

4)

Explaining : A dipole is placed in an electric field as shown. In which direction will it move?

Concept : This question is based on dipole placed in non-uniform electric field.

Solution :

A dipole in a non-uniform electric field experiences a net force that moves it towards the region where the electric field is stronger. The dipole tends to move in a way that decreases its potential energy, which is given by the interaction between the dipole moment and the electric field.

Final Answer : (2)

5) **Explaining :** find force an dipole

Concept : This question is based on $U = -\vec{P} \cdot \vec{E}$

Solution :

$U = -\vec{P} \cdot \vec{E}$ = -P.E

than $F = -\frac{-dU}{dx} = \frac{d(PdE)}{dx}$

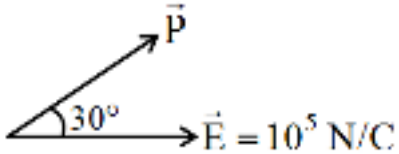
$\frac{dE}{dx} = 1$ Hence F = P

Final Answer : (4)

6) **Explaining :** find PE of dipole

Concept : This question is based on $u = -\vec{P}\vec{E}$

Solution :



τ = PE sin30°

$10\sqrt{3} = \frac{PE}{2}$

PE = $20\sqrt{3}$

Potential energy = – PEcos30°

$= -20\sqrt{3} \times \frac{\sqrt{3}}{2} = -30J$

Final Answer : (3)

7) **Explaining :** To find net resistance increases or decreases.

Concept : Temperature dependence of resistivity.

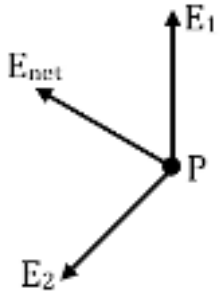
Solution :

$R_1' = R_1(1 + \alpha T)$ $R_2' = R_2(1 - \beta T)$

$\Delta R_1' = \alpha R_1 T$ $\Delta R_2' = -\beta R_2 T$

$R_{eq} = R_1' + R_2'$
$\Delta R_{eq} = \Delta R_1' + \Delta R_2'$
$= T(\alpha R_1 - \beta R_2)$
If $\alpha R_1 > \beta R_2$
$\Delta R_{eq} \rightarrow +ve$
So, net resistance increases with increase in temperature.
Final Answer : (2)

8) **Explaining :** Direction of $\vec{E}$
Concept : This question is based on direction of electric field
Solution :



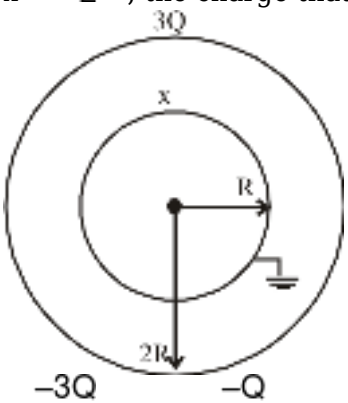
Final Answer : (2)

9) **Explaining :** An α-particle of mass 6.4×10^{-27} kg and charge 3.2×10^{-19} C is situated in a uniform electric field of 1.6×10^5 Vm^{-1}. The velocity of the particle at the end of 2×10^{-2} m path covered, when its starts from rest is
Concept : This question is based on $a = \frac{qE}{m}$
Solution :
$a = \frac{qE}{m} = \frac{3.2 \times 10^{-19} \times 1.6 \times 10^5}{6.4 \times 10^{-27}} = 8 \times 10^{12}$ m/s^2
$\Rightarrow v^2 = 4^2 + 2as$
$\Rightarrow v^2 = 0 + 2 \times 8 \times 10^{12} \times 2 \times 10^{-2}$
$\Rightarrow v = 4\sqrt{2} \times 10^5$ m/s
Final Answer : (4)

10) **Explaining :** potential difference
Concept : This question is based on v = Ed
Solution :
Points 2 and 3 are equipotential points. Hence, potential difference between points 1 and 2 is the same as that between 1 and 3. V = Ed cos 60°
Final Answer : (2)

11) **Explaining :** Property of conductor
Concept : This question is based on v = 0
Solution :
When inner shell is grounded the potential of inner shell becomes zero because potential of the Earth is taken to be zero.
$\frac{Kx}{R} + \frac{K3Q}{2R} = 0$

$x = \frac{-3Q}{2}$, the charge that has increased



$= \frac{-3Q}{2} - (-Q) = \frac{-Q}{2}$.

Hence the amount of charge that flows into the Earth = $\frac{Q}{2}$

Final Answer : (2)

12) **Explaining :** find $\vec{E}$

Concept : This question is based on $E = -\frac{dv}{dr}$

Solution :

$E_x = -(2xy - 1)\,\hat{i} = \hat{i}$

$E_y = -(x^2 - 2y + 2)\,\hat{j} = -2\hat{j}$

Final Answer : (3)

13) **Explaining :** motion of charge particle

Concept : This question is based on $t = \sqrt{\frac{2s}{a}}$

Solution :

For both the charges distance is d/2 and Electric field will be $E = \sigma/\varepsilon_0$ for uniformly accelerated motion –

$s = ut + \frac{1}{2}at^2$

$\Rightarrow t = \sqrt{\frac{2s}{a}}$ (for $u = 0$)

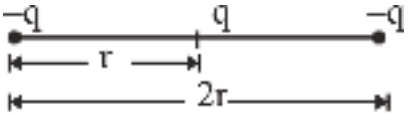
$\frac{t_A}{t_B} = \sqrt{\frac{a_B}{a_A}} = \sqrt{\frac{qE}{2m} \times \frac{m}{2qE}} = \frac{1}{2}$

Final Answer : (1)

14) **Explaining :** find ratio of PE

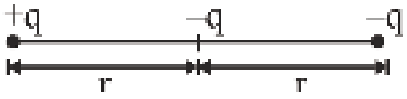
Concept : This question is based on $U = \frac{KQ_1Q_2}{r}$

Solution :



$$U_1 = -\frac{2Kq^2}{r} + \frac{Kq^2}{2r}$$

$$U_1 = \frac{-3}{2}\frac{kq^2}{r}$$

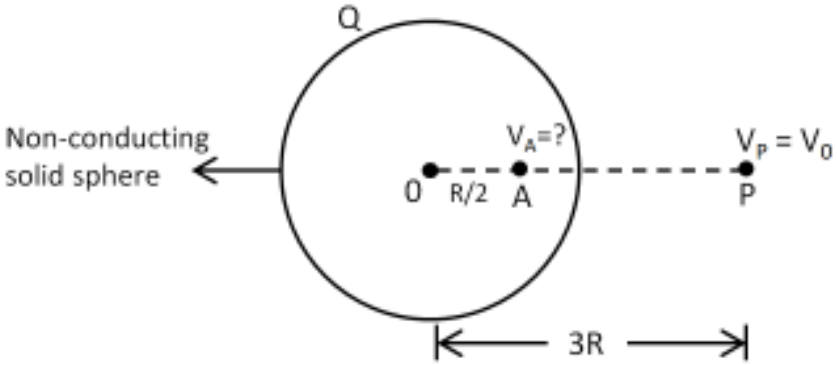


$$U_2 = -\frac{kq^2}{2r}$$

$$\frac{U_1}{U_2} = \frac{3}{1}$$

Final Answer : (1)

15) **Explain Question:**



We have to calculate potential at point A [at distance R/2 from centre] in form of 'V_0'.

Concept:

$$V_{inside} = \frac{KQ}{2R^3}\left(3R^2 - r^2\right)$$ [for solid sphere]

$$V_{outside} = \frac{KQ}{r}$$

Solution:

$$V_P = V_0 = \frac{KQ}{3R}$$

$$V_A = \frac{KQ}{2R^3}\left[3R^2 - \frac{R^2}{4}\right]$$

$$V_A = \frac{KQ}{2R^3}\left[\frac{11R^2}{4}\right]$$

$$V_A = \frac{11}{8}\frac{KQ}{R}$$

$$\frac{V_A}{V_0} = \frac{\frac{11}{8}\frac{KQ}{R}}{\frac{KQ}{3R}}$$

$$\Rightarrow \frac{V_A}{V_0} = \frac{33}{8}$$

$$\Rightarrow V_A = \frac{33}{8}V_0$$

Final Answer: Option (2)

16) **Explaining :** A long wire of linear charge density λ is passed through the centre of

spherical surface of radius R then flux through the surface is :-

Concept : This question is based on $\phi = \frac{q_{en}}{\varepsilon_0}$.

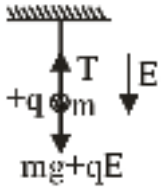
Solution : $\phi = \frac{q_{en}}{\varepsilon_0} = \frac{2R\lambda}{\varepsilon_0}$

Final Answer : (2)

17) **Explaining :** find tension

Concept : This question is based on T = qE + mg

Solution :



T = qE + mg $= (10\times10^{-6})\,(300) + 0.5 \times 10^{-3} \times 10$

$= 8 \times10^{-3}$ N **Final Answer : (2)**

18) **Explaining :** find ratio of $\frac{C}{C_0}$

Concept : This question is based on $C = \frac{A\in_0}{d - t + \frac{t}{K}}$

Solution :

$C_0 = \frac{A\in_0}{d}$

$C = \frac{A\in_0}{d - \frac{d}{4} + \frac{d}{4K}} = \frac{A\in_0(4K)}{d(3K+1)}$

$\therefore \frac{C}{C_0} = \frac{4K}{3K+1}$

Final Answer : (2)

19) **Explaining :** Two charged capacitors have their outer plates fixed and inner plates connected by a spring of force constant 'k'. The charge on each capacitor is q and area of each plate is A. Find the extension in the spring at equilibrium

Concept : This question is based on $kx = \frac{q^2}{2\in_0 A}$

Solution :

$kx = \frac{q^2}{2\in_0 A}$

$x = \frac{q^2}{2\in_0 Ak}$

Final Answer : (1)

20) **Explaining :** In the given circuit a charge of +80μC is given to the upper plate of 4μF capacitor. Then in the steady state, the charge on upper plate of the 3μF capacitor is :

Concept : This question is based on $Q \propto C$

Solution :

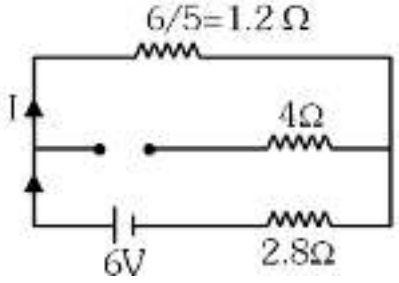
$$Q_1 = \left(\frac{3}{3+2}\right) \times 80 = 48\,\mu C$$

Final Answer : (3)

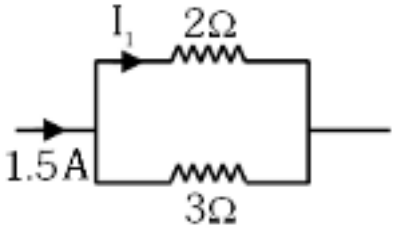
21) **Explaining :** In the figure shown, the capacitor of the capacity C is 2μF. The current in 2Ω resistor is :

Concept : This question is based on ohm law

Solution :



$$I = \frac{V}{R_{eq}} = \frac{6}{4} = \frac{3}{2}A$$



$$I_1 = \frac{3}{5} \times 1.5A = 0.9A$$

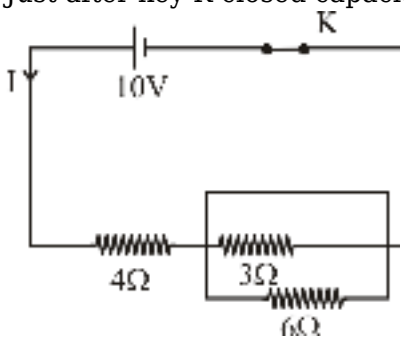
Final Answer : (2)

22) **Explaining :** Determine the current through the battery in the circuit shown, Immediately after the key K is closed.

Concept : This question is based on ohm law

Solution :

Just after key K closed capacitor behave as short circuit so circuit become



$R_{eq} = 4\Omega$

$I = \frac{10}{4}$ A ,

I = 2.5 A **Final Answer : (4)**

23) **Explaining :** Find $V_A - V_B$:-

Concept : This question is based on KVL

Solution :

$$V_A - 6 - 3 \times 2 + \frac{9}{1} - 3 \times 3 = V_3$$

$$V_A - V_B = +12\text{volt}$$

Final Answer : (1)

24) **Explaining :** A light bulb, a capacitor and a battery are connected together as shown here, with switch S initially open. When the switch S is closed, which one of the following is true
Concept : This question is based on at t = 0 capacitor behave as a wire
Solution :
The bulb will light up for an instant when the capacitor starts charging.
Final Answer : (1)

25) **Explaining :** Eight equally charged identical drops combine to form a big drop. If potential of each small drop is 10V, then potential of bigger drop is :
Concept : This question is based on $V = \frac{Kq}{r}$
Solution :

$$V_1 = \frac{Kq}{r} = 10\,V$$

$$V_2 = \frac{KQ}{R} = \frac{K\,8q}{2r}$$

$$= \frac{4\,Kq}{r}$$

$$= 4 \times 10\ V$$

$$= 40\ V$$

Final Answer : (1)

26) **Explaining :** ratio of energy
Concept : This question is based on $U = \frac{1}{2}CV^2$
Solution :
Initially potential difference across both the capacitor is same hence energy of the system is

$$U_1 = \frac{1}{2}CV^2 + \frac{1}{2}CV^2 = CV^2 \quad \text{...(i)}$$

In the second case when key K is opened and dielectric medium is filled between the plates, capacitance of both the capacitors becomes 3C, while potential difference across A is V and potential difference across B is $\frac{V}{3}$ hence energy of the system now is

$$U_2 = \frac{1}{2}(3C)V^2 + \frac{1}{2}(3C)\left(\frac{V}{3}\right)^2 = \frac{10}{6}CV^2 \quad \text{...(ii)}$$

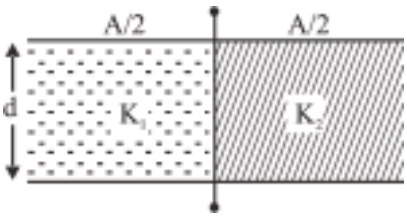
So, $\frac{U_1}{U_2} = \frac{3}{5}$
Final Answer : (3)

27) **Question Explanation:**
An isolated capacitor with charge Q and voltage V has a dielectric (K = 2) inserted. We need to match resulting changes in its properties.
Concept:
Introducing a dielectric increases capacitance. In an isolated system, charge is conserved, infulencing voltage and stored energy based on the new capacitance.
Formula:

$C = \varepsilon A/D$
$C^t = K\varepsilon A/d = KC$
$Q = CV$
$U = \frac{1}{2}QV$
Calculation:
Charge (a): Remains Q (q).
Potential (b): V' = Q/C' = Q/(2C) = V/2(p).
Energy (c): $U' = \frac{1}{2}QV' = \frac{1}{2}Q(V/2) = U/2(p)$.
Capacitance (d): C' = 2C (r).
matching gives a-1, b-p, c-p, d-r, which corresponds to option (1).

28) **Explaining :** If the plates of a parallel plate capacitor connected to a battery are moved apart from each other, then
Concept : This question is based on Q = CV
Solution :
Battery connected so V = constant
Now $d \uparrow \Rightarrow C \downarrow$
(A) : $Q = CV \propto C \Rightarrow Q \downarrow$
(B) : $U = \frac{1}{2}CV^2 \propto C \Rightarrow U \downarrow$
(C) : $C = \frac{\in_0 A}{d} \Rightarrow C \downarrow$
(D) : $\frac{Q}{V} = C \Rightarrow C \downarrow$
(E) : $(Q)(V) \propto C \Rightarrow QV \downarrow$
Therefore statements B, C and E are correct.
Final Answer : (1)

29) **Explaining :** A dielectric slab of dielectric constant K is placed between the plates then compare electric fields E_1, E_2 and E_3.
Concept : This question is based on $E^1 = \frac{E}{K}$
Solution :
$E_2 = \frac{E}{K}$
Final Answer : (2)

30) **Question Explanation:** We need to find capacitance of the system shown in figure.
Concept : Parallel plate capacitor (with dielectric) and parallel combination of capacitors.
Solution :


It is parallel combination of capacitors.

$\Rightarrow C^1 = C_1 + C_2$

$$C^1 = \frac{A}{(2)}\frac{\varepsilon_0 k_1}{d} + \frac{A}{(2)}\frac{\varepsilon_0}{d}k_2$$

$$\boxed{C^1 = \frac{A\varepsilon_0}{(2)\,d}(k_1 + k_2)}$$

Final Answer : option (2). $\frac{\varepsilon_0 A}{d}\left(\frac{K_1 + K_2}{2}\right)$

31) **Explaining :** Two insulated conducting charged spheres placed far away of radii R_1 and R_2 having charges Q_1 and Q_2 respectively are connected to each other, then there is :-

Concept : This question is based on $Q_1R_2 = Q_2R_1$

No loss in energy

Solution :

If $V_1 = V_2$

$$\frac{Q_1}{R_1} = \frac{Q_2}{R_2}$$

$\Rightarrow Q_1R_2 = Q_2R_1$

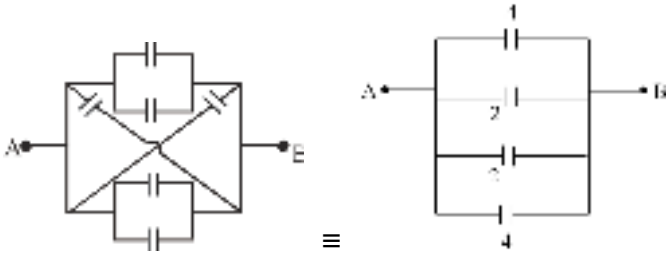
then there is no loss of energy !

Final Answer : ()

32) **Explaining :** Six capacitors each of capacitance of 2μF are connected as shown in the figure. The effective capacitance between A and B is :-

Concept : This question is based on combination of capacitors

Solution :



$C_{AB} = 12\mu F$

Final Answer : (1)

33) **Question**

To find energy density

Concept

energy density of parallel plate capacitance

Formula

$$\text{energy density} = \frac{1}{2}\varepsilon_0 E^2$$

where = E =electric field

$E = V \times d$

V = potential difference

Calculation

$\varepsilon_0 = 8.85 \times 10^{-12}$ S.I unit

$$E = \frac{V}{d} = \frac{300}{2 \times 10^{-3}}$$

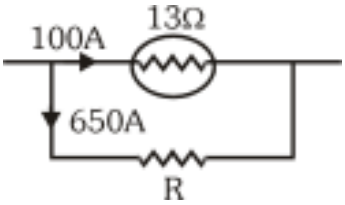
$$\text{Energy density} = \frac{1}{2}\varepsilon_0 E^2$$

$$= \frac{1}{2} \times 8.85 \times 10^{-12} \times \frac{300 \times 300}{4 \times 10^{-6}}$$

$$= 0.095 \approx 0.1 \text{ J/m}^3$$

Conclusion
Hence, option (2) is correct

34) **Explaining :** The resistance of an ammeter is 13 Ω and its scale is graduated for currents upto 100 A. After an additional shunt is connected to this ammeter it becomes possible to measure currents upto 750 amperes by this meter. The value of shunt-resistance is :–
Concept : This question is based on shunt
Solution :



$13 \times 100 = 650 \times R$
$\Rightarrow R = 2\Omega$
Final Answer : (3)

35) **Explaining :** To find the electric field at point P.
Concept : Gauss law
Solution :

$$\in_0 E.4\pi.4R^2 = \frac{Q}{\frac{4}{3}\pi(3R)^3 - \frac{4}{3}\pi R^3}.\left(\frac{4}{3}\pi(2R)^3 - \frac{4}{3}\pi R^3\right)$$

$$\in_0 E.16\pi R^2 = \frac{Q}{\frac{4}{3}\pi 26R^3}.7R^3 \times \frac{4}{3}\pi$$

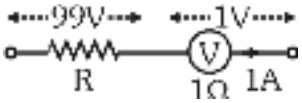
$$\in_0 E = \frac{7Q}{26 \times 16\pi R^2} \qquad E = \frac{7Q}{416\pi \in_0 R^2}$$

Final Answer : (1)

36) **Explain Question:**
Question is asking value of resistance which can increase range of voltmeter to 100 V.

Concept:
This question is based on voltmeter

Solution:



$99 = 1 \times R \Rightarrow R = 99\Omega$

Final Answer: (3)
99 Ω

37) **Explanation :** In mis question we have to tell which fell which bulb will glow brightest in the given circuit diagram if all bulbs are identical.
Concept :- Electric power :-
Formula :- $P = I^2R$
Calculation :- As all bulls are identical and the power (related by brightness) is given as $P = I^2R$ so me bulb with the most current will be brightest. In Parallel connection, the current is spilt between the branches while in series the current will be same.
So in figure we can set mat bulb 'C' will receive into maximum current while the bulbs A, B, D will not get complete current as 'C' as ***m*** are connected in Parallel ***trenches***. So bulb 'C' will be the brightest.

Conclusion : The current answer option (3).

38) **Explaining :** Find the power dissipated in the circuit.
Concept : This question is based on wheat stone Bridge
Solution :
Its a balanced WSB.

$$R_{eq} = \frac{(8)(4)}{8+4} = \frac{8}{3}\Omega$$

$$P = \frac{V^2}{R_{eq}} = \frac{(2)^2}{\frac{8}{3}} = \frac{3}{2}W = 1.5W$$

Final Answer : (1)

39) **Explaining :** In a Wheatstone bridge shown in figure, x = y and A > B. The direction of the current between a and b will be
Concept : This question is based on wheat stone Bridge
Solution :
Theoretical
Final Answer : (2)

40) **Explaining :** To predict the value of $T_2 - T_1$
Concept : Temp. dependance of resistivity.
Solution :
$R_1 = \tan\theta \quad R_2 = \tan(90-\theta) = \cot\theta$
$R_1 = R_0(1 + \alpha T_1) \quad R_2 = R_0(1 + \alpha T_2)$
$R_2 - R_1 = R_0\,\alpha\,(T_2 - T_1)$
$\cot\theta - \tan\theta\ \alpha\ (T_2 - T_1)$

$$T_2 - T_1\ \alpha\ \frac{2(\cos^2\theta - \sin^2\theta)}{2\cos\theta\sin\theta}$$

$$T_2 - T_1\ \alpha\ \frac{2\cos 2\theta}{\sin 2\theta} \propto \cot(2\theta)$$

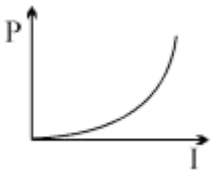
Final Answer : (3)

41) **Explaining :** In given meter bridge circuit length of wire AB is 1m. When resistance X is shunted by R resistance then balancing length from A is found to be 20 cm. Find value of R :-
Concept : This question is based on meter bridge
Solution :
$$\frac{\frac{6R}{6+R}}{3} = \frac{20}{80} = \frac{1}{4}$$
$$\frac{6R}{6+R} = \frac{3}{4} \Rightarrow R = \frac{6}{7}\Omega$$
Final Answer : (1)

42) **Explaining :** 100 cells, each of e.m.f. 5V and internal resistance 1Ω, are to be arranged so as to drive maximum current in a 25Ω resistance. Each row is to contain equal number of cells. The number of rows should be: -
Concept : This question is based on cell
Solution :
Let number of coloum - n and
number of row - m
for I_{max}
$$\frac{nr}{m} = R$$
$$\frac{n \times 1}{m} = 25 \quad \ldots(1)$$
and nm = 100 ...(2)
solve eq. (1) and (2)
m = 2
Final Answer : (1)

43)

Explain Question :- The question asks to identify the graph of power versus current from a given current versus voltage graph for a non-ohmic device.
Concept :-This question is based on Relationship between power, voltage, and current (P = VI)
Solution :-
$V \propto I^2$
using power formula
$P \propto VI \propto I^3$



Final answer :-(2)

44) **Explaining :** The area of cross section of current carrying conductor is A^0 and $\frac{A_0}{4}$ in section (1) & (2) if v_{d1}, v_{d2} and E_1, E_2 are drift velocity and electric field at sec. (1) and (2) respectively, then :-
Concept : This question is based on $V \propto \frac{1}{A}$

Solution :

$$\frac{v_1}{v_2} = \frac{1}{4}$$

$$J = \sigma E \Rightarrow E \propto \frac{1}{A}$$

$$\frac{E_1}{E_2} = \frac{4}{1}$$

Final Answer : (4)

45) **Explaining :** Two wires A and B of equal masses and of the same metal are taken. The diameter of the wire A is half the diameter of the wire B. If the resistance of the wire A be 24 ohm, the resistance of the wire B is :-

Concept : This question is based on $R = \frac{\rho \ell}{A}$

Solution :

$m_A = m_B$

$\left(\frac{\pi r^2}{4}\right) \ell_1 d = (\pi r^2) \ell_2 d$

$\Rightarrow \frac{\ell_1}{\ell_2} = 4$ ..... (1)

Now $\frac{R_1}{R_2} = \frac{\ell_1}{\ell_2} \times \left(\frac{r_2}{r_1}\right)^2$

$= 4 \times \left(\frac{r}{r/2}\right)^2$

$= 16$

$\Rightarrow R_2 = \frac{24}{16} = 1.5\ \Omega$

Final Answer : (2)

CHEMISTRY

46) **Solution :**

$CuSO_4.5H_2O$ is $[Cu(H_2O)_4]SO_4.H_2O$

4, water molecules are attached as secondary valency and 1, water molecule is attached due to H-bond.

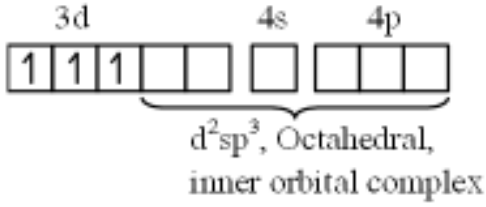
Final Answer : (2)

47) **Solution :**

$[Cr^{+3}(NH_3)_6]Cl_3 + AgNO_3 \rightarrow AgCl\downarrow$

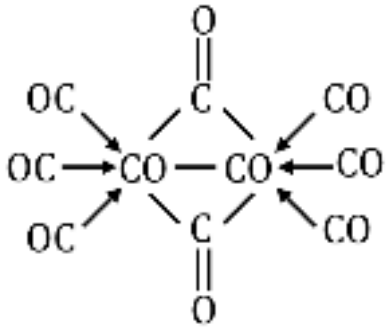
white

$Cr^{+3} = [Ar]4s^0\ 3d^3$



Final Answer : (3)

48) **Solution :**



Final Answer : (2)

49) **Solution :** $CoCl_3.5NH_3 \rightarrow [Co(NH_3)_5Cl]Cl_2$
Final Answer : (2)

50) **Solution :**
$H_2O \rightarrow$ W.F.L.
$CN^- \rightarrow$ S.F.L.
Final Answer : (3)

51) **Solution :** As per CFT
Final Answer : (3)

52) **Solution :** $[RhCl(PPh_3)_3]$
Final Answer : (4)

53) **Solution :** H_2O will work as S.F.L.
Final Answer : (4)

54) **Solution :** Strong field ligand & Weak field ligand
Final Answer : (3)

55) **Solution :** $[Ni(CO)_4]$, $[NiCl_4]^{-2}$
Final Answer : (3)

56) **Solution :** $[Pt(NH_3)_4Cl_2]Cl_2$, $[Ni(NH_3)_6]Cl_2$
Final Answer : (4)

57) **Solution :** Oxidation state = +3
Final Answer : (2)

58) **Solution :** As per rules.
Final Answer : (3)

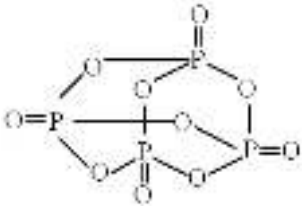
59) **Solution :**
$\left[\overset{+3}{Cr}(NH_3)_2Cl_2(C_2O_4)\right]^-$ $[CrCl_6]^{-3}$
Co-ordination no. = 6 Charge =–3
Final Answer : (2)

60) **Solution :** $[Co(NH_3)_5Br]SO_4$ forms ppt. of $BaSO_4$ with $BaCl_2$
Final Answer : (4)

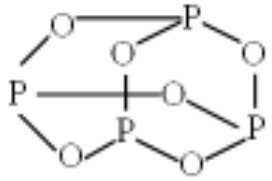
61) **Solution :** F^- → W.F.L.
Final Answer : (1)

62) **Solution :** Tetrahedral geometry
Final Answer : (4)

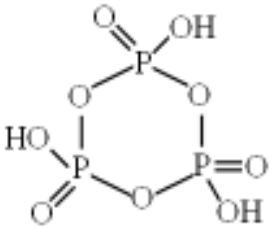
63) P_4O_{10}



P_4O_6



$H_3P_3O_9$



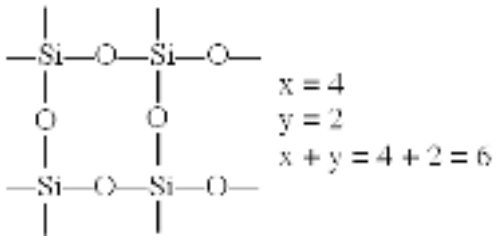
64) Order of bond angle :
$NH_3 > NF_3$
$PH_3 < PF_3$
$AsH_3 < AsF_3$
$AsF_3 < AsCl_3$
$H_2O > F_2O$

65) Hydrolysis of SF_6 is not possible.

66) Due to back bonding $(SiH_3)_3N$ is not a electron pair donor.

67) Correct order of IE is,
He > Ne > Ar > Kr > Xe

68) HBr is stronger acid than HCl

69) 

70)

Inert pair effect.

71)

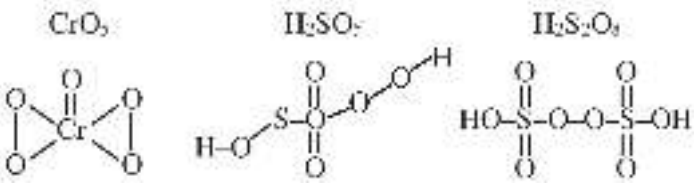
Mg_2C_3 gives propyne on hydrolysis.

72)

Boric acid is polymeric due to the presence of hydrogen bonds

73)

Has carbon atoms arranged in large plated of rings of strongly bonded carbon atoms with weak interplate bonds

74) 

75)

$Br_2 + F^- \rightarrow Br^- + F_2$ reaction not possible

order of oxidising → $F_2 > Cl_2 > Br_2 > I_2$

76)

$Cl_2 > Br_2 > F_2 > I_2$

77)

Si > C > P > N

78) BBr_3 → trigonal planner
TlO_2 → Basic
$B(OH)_3$ → Monobasic
$AlCl_3$ → Dimer

79) Down the group stability of +4 oxidation state decrease.

80) (2p–2p) π bonding is effective whereas.
(3p–2p) π bonding is not effective.

81) **Solution :** Transition element
Final Answer : (4)

82) **Solution :** Os
Final Answer : (3)

83) **Solution :** Sc > Ti > V > Cr > Mn > Fe ≈ Co ≈ Ni < Cu < Zn
Final Answer : (2)

84) **Solution :** Lanthanoid contraction.
Final Answer : (3)

85) **Solution :** I^- is a better reducing agent.
Final Answer : (3)

86) **Solution :**
In $Cr_2O_7^{-2}$, six Cr–O bonds have partial double bond character while two Cr–O bonds have single bond character.
Final Answer : (3)

87) **Solution :**
(1) Passing CO_2 gas
(2) By passing Cl_2
(3) Electrolytic oxidation

Final Answer : (4)

88) **Solution :** Copper and zinc
Final Answer : (2)

89) **Solution :** Cr^{+6} → O.S.
Final Answer : (4)

90) **Solution :** d-block elements can form interstitial compounds.
Final Answer : (4)

BIOLOGY

91) **Solution :** In some prokaryotes like cyanobacteria or Blue-Green algae, there are other membranous extensions in the cytoplasm called chromatophores.
Final Answer : (2)

92) **Solution :** Ribosome is sub-spherical granular structure which is devoid of any covering membrane.
Final Answer : (2)

93) **Solution :** The ratio of protein and lipids varies considerably in different cell types. The membrane of the erythrocytes has approximately 52% of proteins and 40% lipids.
Final Answer : (2)

94) **Solution :** RBC (red blood cell) of the human is the best material for the study of structure of cell membrane. Chemical studies on the RBC cell membrane enabled the scientists to infer the possible structure of plasma membrane. The plasma membrane of the RBC consists of a complex, well-ordered group of lipids and proteins stretched over the outer surface of the cell in the form of a lipid bilayer punctuated by penetrating or attached proteins. This membrane has numerous properties that arise in part from specialized interactions between specific membrane proteins or lipids, or both.
Final Answer : (1)

95) **Solution :** Plasmodesmata is an effective transport pathway present between two adjacent cells. They are microscopic channels which traverse the cell walls of plant cells and some algal cells, enabling transport and communication between them.
Final Answer : (1)

96) **Solution :** Plastids found in all plant cells and in euglenoides are called elaioplast which stores fats.
Final Answer : (1)

97) **Solution :** Axoneme with 9 + 2 microtubular arrangement occurs in cilia and flagella. Cilia are hair like growths of the cell membrane. Flagella are comparatively longer and responsible for movement. The axoneme usually has nine pair of peripherally located microtubules. Such an arrangement of axonemal microtubules is referred to as the 9 + 2 array.
Final Answer : (3)

98) **Solution :** Ribosomes are about 15 nm by 20 nm in size.
Final Answer : (1)

99) **Solution :** All the given statements regarding cilia and flagella are correct.
Final Answer : (2)

100) **Solution :** In bacteria, a sticky, gelatinous material called glycocalyx collects outside the cell wall to form an additional surface layer. Glycocalyx covering of the bacterial cell may be loose, thin sheath or thick and tough coat. Loose glycocalyx covering is termed slime layer while tough and thick type is known as capsule.
Final Answer : (3)

101) **Solution :** Endomembrane system is a grouping of some membrane organelles which function in close coordination with one another, viz., endoplasmic reticulum, Golgi complex, lysosomes and vacuoles.
Final Answer : (1)

102) **Solution :** All the functions described are performed by Golgi apparatus. The Golgi apparatus is an organelle found in most eukaryotic cells. It was identified in 1897 by the Italian physician Camillo Golgi and named after him in 1898. Golgi apparatus is the important site of formation of glycoproteins and glycolipids.
Final Answer : (4)

103) **Solution :** In animal cells lipid like steroidal hormones are synthesized in SER. In plant cells, the vacuoles can occupy up to 90% of the volume of the cell. Chlorophyll and carotenoid pigments are responsible for trapping light energy essential for photosynthesis.
Final Answer : (1)

104) **Solution :** The centrosome (occur only in animals cells) is the main place where cell microtubules get organized. Centrosome usually contains two cylindrical structures called centrioles. Centrioles are composed of groupings of microtubules arranged in a 9 + 3 pattern. The pattern is so named because a ring of nine microtubule "triplets" are arranged at right angles to one another. Centrioles, found in animal cells, help to organize the assembly of microtubules during cell division. Centrioles replicate during the interphase stage of mitosis and meiosis.
Final Answer : (1)

105) **Solution :** Based on the position of the centromere, the chromosomes can be classified into four types: metacentric (centromere middle), sub-metacentric (centromere slightly away

from middle), acrocentric (centromere close to end) and telocentric (terminal centromere). The metacentric chromosome has middle centromere forming two equal arms of the chromosome. In acrocentric chromosome the centromere is situated close to its end forming one extremely short and one very long arm.
Final Answer : (4)

106) **Solution :** Leeuwenhoek – He is commonly known as "the Father of Microbiology", and considered to be the first microbiologist. First saw and described a living cell. Robert Brown was a Scottish botanist and paleobotanist who made important contributions nucleus and cytoplasmic streaming. Schleiden was a German botanist and co-founder of the cell theory, along with Theodor Schwann and Rudolf Virchow. He discovered that all plants are composed of different kind of cells. Schwann - He discovered that presence of cell wall is unique to plant cells.
Final Answer : (1)

107) **Solution :** A – IV; B – III; C – II; D – I
Viruses — 0.02 - 0.2 micrometer
PPLO (Pleuro Pneumonia like Organisms) — About 0.1 micrometer
Eukaryotic cells — 10-20 micrometer
Bacterium — 1-2 micrometer
Final Answer : (2)

108) **Solution :** Tonoplast is a membrane separating a vacuole from the surrounding cytoplasm in a plant cell. It controls ionic movement in and around the cell. Contractile vacuole is a sub-cellular structure found in unicellular algae. It absorbs excess water and wastes from a microorganism’s cell and excretes them into the environment. Food vacuole is a membrane-unclosed cell vacuole with a digestive function. It contains material taken up in by the process of phagocytosis. Air vacuole is a small cavity or space in the tissues. It stores metabolic gases.
Final Answer : (1)

109) **Solution :** Plasmodesmata are cytoplasmic bridges between two adjacent cells which connect the neighboring cells.
Membrane surrounding the vacuole in plants is known as Tonoplast.
Final Answer : (2)

110) **Solution :** Centrioles serve as basal bodies for cilia and flagella. They are concerned with spindle formation during cell division. Lysosomes are membrane-bound organelles containing many hydrolytic enzymes, which are optimally active at an acid pH (near pH 5). Thylakoids are membrane-bound organelles found within chloroplasts. The thylakoid membrane, forms many flattened, fluid-filled tubules that enclose a single convoluted compartment. These tubules tend to stack on top of each other to form a structure called a granum.
Final Answer : (3)

111) **Solution :** In the given figure of cilia/flagella, the marked part as A, B, C and D are respectively plasma membrane, inter doublet bridge, central microtubule, and radial spoke.

Final Answer : (1)

112) **Solution :** In the given diagram of chloroplast, the parts marked as A, B, C and D are respectively granum, thylakoid, stromal lamella and stroma. Granum is a stacked membranous structure that contains the chlorophyll and is the site of the light reactions of photosynthesis. Thylakoids are saclike membranes that make up grana. Stromal lamella are the extensions that run from one granum, through the stroma, into a neighboring granum. The space between the inner membrane and the thylakoid membrane is filled with stroma, a matrix containing dissolved enzymes, starch granules, and copies of the chloroplast genome.
Final Answer : (2)

113) **Solution :** In the given diagram of mitochondria, the structure marked as A, B, C, D and E are respectively outer membrane, inner membrane, matrix (space within the inner membrane), intermembrane space (the space between the outer and inner membranes) and crista (formed by infoldings' of the inner membrane). Mitochondria are a membrane-bound organelle found in most eukaryotic cells and also called as the power house of the cell.
Final Answer : (1)

114) **Solution :** The given diagram shows the fluid mosaic model of cell membrane. In the diagram the components marked as A, B, C, D and E are respectively (i) sugar, (ii) protein, (iii) lipid bilayer, (iv) integral proteins, and (v) cytoplasm. The fluid mosaic model was devised by SJ Singer and GL Nicolson in 1972, describes the cell membrane as a two-dimensional liquid in which phospholipid and protein molecules diffuse easily.
Final Answer : (1)

115) **Solution :** The figure shows the structure of metacentric (3) chromosomes. The parts marked as A and B are respectively are satellite and secondary constriction. Metacentric chromosomes have the centromere in the centre, due to which both sections are of equal length. Human chromosome 1 and 3 are metacentric.
Final Answer : (2)

116) **Solution :** Lysosomes are membrane-enclosed organelles and consist of hydrolytic enzymes which capable of breaking down all types of biological polymers (like proteins, nucleic acids, carbohydrates, and lipids). If lysosomes get ruptured in a cell, the cell dies due to release of hydrolytic enzymes which destroys the cells.
Final Answer : (1)

117) **Solution :** Simple diffusion is a process in which plasma membrane does not require any carrier molecule to pass small non- charged molecules or lipid soluble molecules between the phospholipids to enter or leave the cell. The particles move from areas of high concentration to areas of low concentration (they move down their concentration gradient). Oxygen and carbon dioxide and most lipids enter and leave cells by simple diffusion.
Final Answer : (3)

118) **Solution :** Vacuole is a non - living reservoir, bounded by a selectively permeable membrane, the tonoplast. It is not a air filled cavity but it is filled with a highly concentrated

solution called vascular sap or cell sap. pH of vacuolar cell sap is acidic and hypertonic.
Final Answer : (3)

119) **Solution :** Cell organelles like ribosomes and nucleolus lack the unit membrane. Ribosomes are found individually in the cytoplasm and also line the membranes of the rough endoplasmic reticulum. It is composed of RNA and ribosomal protein. Nucleolus is a part of nucleus and characterized by the absence of limiting membrane, presence of chromatin and granules and fibrils of RNA and proteins.
Final Answer : (3)

120) **Solution :** We can eliminate bacteria because they lack organelles completely. Of the organelles listed the only unique one is the chloroplast, which is found exclusively in plants.
Final Answer : (4)

121) **Solution :** G_0 phase is non-cycling quiescent stage of cell which leaves the cell cycle in the early half of G_1 phase due to absence of cycling factors.
Final Answer : (2)

122) **Solution :** Prophase is the stage of mitosis in which the following events takes place like,
(i) The nucleolus disappears.
(ii) In the cytoplasm, the mitotic spindle, consisting of microtubules and other proteins, forms between the two pairs of centrioles as they migrate to opposite poles of the cell.
(iii) The nuclear envelope disappears at the end of prophase. Which indicates the beginning of the sub stage called prometaphase.
(iv) Chromosomes become more coiled and can be viewed under a light microscope.
(v) Each duplicated chromosome is seen as a pair of sister chromatids joined by the duplicated but unseparated centromere.
Final Answer : (2)

123) **Solution :** Meiosis I is known as reductional division due to reduction in the number of chromosomes. Meiosis II is called equational division because of maintaining the same number of chromosomes.
Final Answer : (1)

124) **Solution :** The process of pairing of homologous chromosomes is called synapsis. Synapsis occurs during zygotene stage of meiosis-I.
Final Answer : (4)

125) **Solution :** The synaptonemal complex is formed at the region of synapsis. It is a protein structure that forms between homologous chromosomes (two pairs of sister chromatids) during meiosis and is thought to mediate chromosome pairing, synapsis, and recombination. This stage appears in zygotene stage of prophase I of meiosis I.
Final Answer : (4)

126) **Solution :** Leptotene is the first stage of prophase I of meiosis I. Leptotene is of very short duration in which progressive condensation and coiling of chromosomal fibres takes place. In leptotene, the nuclear membrane enlarges and chromosomes become distinct. Chromosomes appear as long and slender threads bearing beads like structure called chromomeres.
Final Answer : (2)

127) **Solution :** A homologous chromosome is a set of one maternal chromosome and one paternal chromosome that pair up with each other inside a cell during meiosis.
Final Answer : (1)

128) **Solution :** In plant cells, wall formation starts from the centre of the cell and grows outward to meet the existing lateral walls (centrifugal).
Final Answer : (2)

129) **Solution :** Terminalization of chiasmata occurs in diakinesis stage of meiosis I. It is the disappearance of chiasmata of sliding towards the tips of the chromosomes due to tight condensation.
Final Answer : (2)

130) **Solution :** All the given statements are correct.
Final Answer : (4)

131) **Solution :** During S or synthesis phase, replication or duplication of chromosomal DNA and synthesis of histone proteins takes place. If the initial amount of DNA is denoted as 2C then it increases to 4C. However, there is no increase in the chromosome number; if the cell has diploid or 2n number of chromosomes at G_1, even after S phase the number of chromosomes remains the same, i.e., 2n.
Final Answer : (2)

132) **Solution :** All the statements regarding G_o stage of the cell are correct. G_o stage or resting phase is a period in the cell cycle in which cells exist in a quiescent state. G_o phase is viewed as either an extended G_1 phase, where the cell is neither dividing nor preparing to divide, or a distinct quiescent stage that occurs outside of the cell cycle. Some (like cells, nerve and heart muscle cells), become quiescent when they reach maturity (i.e., when they are terminally differentiated) but continue to perform their main functions for the rest of the organism's life.
Final Answer : (4)

133) **Solution :** Crossing over leads to recombination between homologous chromosomes. Recombination is the rearrangement of genetic information within and among DNA molecules.
Final Answer : (2)

134) **Solution :** Diplotene is the longest and most active subphase of prophase I of meiosis. The beginning of diplotene is recognized by the dissolution of the synaptonemal complex and

the tendency of the recombined homologous chromosomes of the bivalents to separate from each other except at the sites of crossovers. These X shaped structures are called chiasmata. Diplotene can last for months and years in oocytes of some vertebrates.
Final Answer : (2)

135) **Solution :** During metaphase discontinuous fibres radiate out from two spindle poles and get connected to the disc shaped structure at the surface of the centromere called kinetochores. These are called chromosome fibres or spindle fibres. A kinetochore is a complex protein structure that is analogous to a ring for the microtubule hook; it is the point where microtubules attach themselves to the chromosome.
Final Answer : (3)

136) **Solution :** Crossing over is the exchange of genetic material between two homologous pair of chromosomes. Crossing over is also an enzyme-mediated process and the enzyme involved is called recombinase.
Final Answer : (3)

137) **Solution :** During Metaphase II the chromosomes align along the equator. The microtubules from opposite poles of the spindle get attached to the kinetochores of sister chromatids. Anaphase II occurs with the simultaneous splitting of the centromere of each chromosome move towards the opposite poles of the cell.
Final Answer : (4)

138) **Solution :** A – IV, B – III, C – II, D – I
Metaphase – Chromosomes are moved to spindle fibre.
Anaphase – Centromere splits and chromatids apart.
Zygotene – Pairing between homologous chromosomes takes place.
Pachytene – Crossing between homologous chromosomes occurs.
Final Answer : (3)

139) **Solution :** Terminalization which occurs in diakinesis stage of meiosis I, is the disappearance of chiasmata of sliding towards the tips of the chromosomes due to tight condensation. Synapsis is the pairing of two homologous chromosomes. It allows matching-up of homologous pairs prior to their segregation, and possible chromosomal crossover between them. The process of attachment of homologous chromosomes leads to the formation of bivalents. Chiasmata is a point at which paired chromosomes remain in contact during the first metaphase of meiosis, and at which crossing over and exchange of genetic material occur between the strands. Synaptonemal complex is formed at the region of synapsis. It is a protein structure that forms between homologous chromosomes (two pairs of sister chromatids) and is thought to mediate chromosome pairing, synapsis, and recombination.
Final Answer : (1)

140) **Solution :** A – II, B – III, C – IV, D – I
Anaphase – Movements of chromatids towards opposite poles.
Prophase – Initiation of assembly of mitotic spindle.
Interphase – Proteins are synthesized in preparation for mitosis while cell growth continues.

Metaphase – Spindle fibres attach to the kinetochores of chromosomes.
Final Answer : (1)

141) **Solution :** In schematic break-up of the phases/ stages of cell cycle, D-synthetic phase is the correct indication of the stage/phase in the cell cycle.
Final Answer : (2)

142) **Solution :** The figure A represents late anaphase while figure B represent prophase stage of mitosis.
Final Answer : (3)

143) **Solution :** The correct identification of X, Y and Z are respectively G_1, S and G_2. During G_1 phase, cell stores ATP for cell division and synthesis of protein, RNA, amino acids, ATP and nucleotides occur. S phase, also called invisible stage of M phase, includes the synthesis of DNA and histone proteins. In G_2 phase synthesis of RNA and proteins continues and formation of macromolecules for spindle and organelle formation occurs. In this the quantity of DNA within the cells has increased to 4C but the cell is still considered diploid.
Final Answer : (1)

144) **Solution :** The given figure shows (I - phase). Interphase (called resting phase) is the time during which cell is preparing for division by undergoing both cell growth and DNA replication in a systemic manner. According to the given figure, the phases marked as A, B, C and D are respectively G_1, S, G_2 and M phase.
G_1 phase: It is also called antephase as during the phase the cell stores ATP for cell division. In this phase, synthesis of protein, RNA, amino acids, ATP and nucleotides occur.
S phase: It is also called invisible stage of M phase. In this stage, synthesis of DNA and histone proteins occurs.
G_2 Phase: In this phase, synthesis of RNA and proteins continues and formation of macromolecules for spindle and organelle formation occurs. In this the quantity of DNA within the cells has increased to 4C but the cell is still considered diploid.
M phase: In M phase, mitosis occurs when the nucleus of the cell divides into two identical nuclei with the same number and type of chromosomes, followed by cytokinesis when the cytoplasm, for both plant and animal cells, divides, thus creating two daughter cells that are genetically equal and approximately identical in size.
Final Answer : (3)

145) **Solution :** One cell produces 4 daughter cells after meiotic division. Thus, for the formation of 100 pollen grains, 100/4 = 25PMC are needed and each will undergo one reduction/meiotic division.
Final Answer : (1)

146) **Solution :** During zygotene and S phase, the formation of bivalents and DNA replication occurs respectively. A bivalent, is a pair of associated homologous chromosomes held together by a complex after chromosome replication. During meiosis the process of synapsis occurs in which bivalents are formed S phase is the synthesis phase during which replication of choromosomes occur.

Final Answer : (4)

147) **Solution :** Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms whereas fertilization restores the diploid phase.
Final Answer : (3)

148) **Solution :** The correct sequence of the events in the meiosis is: III, II, I, IV.
III. Synapsis: Synapsis is the pairing of two homologous chromosomes that occurs during meiosis. It allows matching-up of homologous pairs prior to their segregation, and possible chromosomal crossover between them. It occurs in zygotene sage.
II. Crossing over: Crossing over is exchange of chromosomes segments between non-sister chromatids of homologous pair. Crossing over occurs during pachytene stage.
I. Terminalization: Terminalization of chiasmata occurs in diakinesis stage of meiosis I. It is the disappearance of chiasmata of sliding towards the tips of the chromosomes due to tight condensation.
IV. Disjunction of genomes: The process of breakdown of homologous chromosomes is called disjunction. This process occurs in anaphase I.
Final Answer : (3)

149) **Solution :** During meiosis, daughter cells have the half number of chromosomes with respect to parent cells. Therefore, daughter cell will be genetically different.
Final Answer : (2)

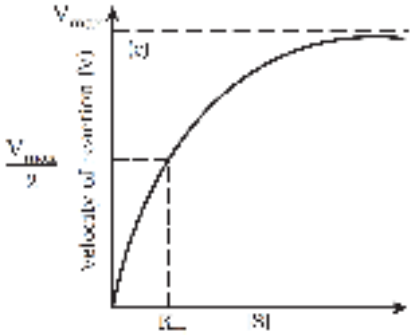
150) **Solution :** In animal cell cytokinesis involves the contraction of the contractile ring of microfilament. Contractile ring of microfilaments develops peripherally below the cell membrane in the equatorial region.
Final Answer : (2)

151) **Solution :** Ribozymes are nucleic acids with catalytic power
Final Answer : (1)

152) **Solution :** Peptide bond is formed between amino group of one amino acid with carboxylic group of another amino acid.
Final Answer : (2)

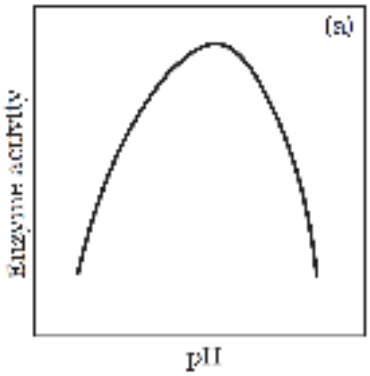
153) **Solution :** The given structure is of cholesterol. It consists of four fused rings.
Final Answer : (3)

154) **Solution :**



Final Answer : (3)

155) **Solution :**



Final Answer : (3)

156) **Solution :** C – C is not catalysed by ligase.
Final Answer : (1)

157) **Solution :** A – IV; B – III; C – II, D – I

A.	Fatty acid	IV.	Palmitic acid
B.	Phospholipid	III.	Lecithin
C.	Aromatic amino acid	II.	Tryptophan
D.	Acidic amino acid	I.	Glutamic acid

Final Answer : (2)

158) **Solution :** A – III; B – IV; C – II; D – I

A.	Collagen	III.	Intercellular ground substance
B.	Trypsin	IV.	Enzyme
C.	Insulin	II.	Hormone
D.	GLUT-4	I.	Glucose transport

Final Answer : (1)

159) **Solution :** A – IV; B – I; C – II; D – III

A.	Enzymes catalysing breakdown without addition of water	IV.	Lyases
B.	Enzyme catalyzes the breakdown with addition of water.	I.	Hydrolases
C.	Enzyme where catalysis involves transfer of electrons.	II.	Oxidoreductase
D.	Enzyme catalysing bonding of two components with the help of ATP.	III.	Ligases

Final Answer : (3)

160) **Solution :** Thymine is a pyrimidine whereas adenine is a purine.
Final Answer : (1)

161) **Solution :** Collagen is the most abundant protein in human body.
Final Answer : (1)

162) **Solution :** Unsaturated fats have both single and double bonds in their structure, which make them metabolically more reactive.
Final Answer : (3)

163) **Solution :** Amino acids are classified either on the basis of their R group or on the basis of synthesis in body.
Final Answer : (3)

164) **Solution :** Proteins are heteropolymer made up of amino acids.
Final Answer : (2)

165) **Solution :** Chitin is a structural polysaccharide.
Final Answer : (1)

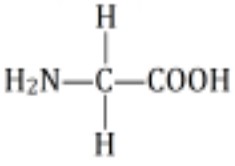
166) **Solution :** Most enzymes are proteins but some are nucleic acids.
Final Answer : (3)

167) **Solution :** Enzymes acts as catalyst by lowering down the activation energy.
Final Answer : (3)

168) **Solution :** Inorganic catalyst work efficiently at high temperature and high pressure.
Final Answer : (4)

169) **Solution :** Co-enzyme is often a vitamin.
Final Answer : (4)

170) **Solution :** The Km value of the enzyme is the value of the substrate concentration at which the reaction reaches to ½ V_{max}.
Final Answer : (3)



171) **Solution :** The simplest amino acid is glycine.
Final Answer : (1)

172) **Solution :** Palmitic acid has only single bond and is a saturated fatty acid. Hence has high M.P.
Final Answer : (4)

173) **Solution :** Trichloroacetic acid is used for preliminary separation of biomolecules in a living tissue.
Final Answer : (3)

174) **Solution :** Valine is a neutral amino acid.
Final Answer : (4)

175) **Solution :** Phospholipids are important cell membrane constituents because they contain polar and non-polar portions.
Final Answer : (4)

176) **Solution :** Lecithin is a phospholipid.
Final Answer : (2)

177) **Solution :** Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of competitive inhibition.
Final Answer : (3)

178) **Solution :** The cofactor of the enzyme carboxypeptidase is zinc.
Final Answer : (1)

179) **Solution :** Substrate binding to active site → Substrate enzyme complex formation → Chemical bonds of the substrate broken → Release of products → Free enzyme ready to bind with another substrate.
Final Answer : (1)

180) **Solution :** Haem is a prosthetic group which acts as a cofactor.
Final Answer : (4)